Blind Watchmaking

Harmonic Latent Spaces and the Geometry of Bias and Variance

Ryan J. Brooks

Abstract

Do you exist in other people's realities the same way you exist in your own? It is a question the field tends to defer: the reflex is to hand the burden of proof back to whoever raised it, as if demanding evidence were the neutral posture and asking was the indulgence. That reflex is not neutral. It is a prior, learned so early in life it passes for common sense. The dominant program in machine learning rests on one such prior: that a single architecture, given enough parameters, data, and compute, can reach anything worth learning - that the architecture itself is neutral ground. This is the Universal Approximation Theorem, and we do not contest it. The theorem is a promise of existence: with enough width, a network can represent the target. It is silent on what follows - which of those representable functions finite training actually reaches, and at what cost. Our claim lives in that silence, alongside the theorem rather than against it: what a model learns in finite training is set less by how large it is than by the geometry of its representation - the shape of the space its weights are allowed to occupy. Approximation says the destination exists; geometry says which destinations are reachable, and how cheaply. We make that concrete with two components of the Praxis framework [8]: a harmonic head that writes a learned standing wave over its features, and a byte-latent encoder [41], which finds that rhythm in content-adaptive byte patches. The same geometry reaches its finest, per-feature form in a plain transformer, so across these approaches it is the granularity that changes, not the geometry. In both, the classical bias-variance tradeoff stops being a single dial. A static, population-average rhythm (the spectrum the field phase-locks to) is pure bias; an input-conditional modulation of that rhythm is structured variance; and we argue the two occupy orthogonal axes of a harmonic latent space, so that they can be lowered together rather than traded against each other - an interpretation of the diagnostics this architecture exposes, offered with its falsifier rather than as a theorem. This reframing motivates a central conjecture: consensus modeling - learning the mean of a corpus - scales linearly in the patterns it must express, while harmonic modeling, which composes patterns by interference, scales logarithmically, because K frequency components address exponentially many configurations. The framework exists so that the reader can test it.

1 Introduction

The dominant recipe for capability is scale. Fix a transformer [58], then grow it: more parameters, more tokens, more compute, with returns described by smooth scaling laws [28]. The recipe works, but it assumes the architecture itself imposes no ceiling on which patterns gradient descent can find - that everything worth learning is reachable from any sufficiently large model. This paper questions that assumption and offers a concrete alternative: that the *geometry* a model is allowed to represent, not its size, determines the patterns accessible to it during finite training.

With geometry as defined (Section 2), the objects in play are *symbols* - discrete geometries the model reuses - and what it learns is a change in symbolism over time. The oldest tradeoff in statistics gives the first coordinate on that change. A model that minimizes error by collapsing toward the average of its training data has high bias and low variance; a model that chases the particulars of each example has low bias and high variance. Classically these trade off along a single line: you choose a point on it. Our first contribution is to show that in a *continuous, harmonic* latent space this is no longer true. Bias and variance become separate coordinates, and a model can move in one without paying in the other.

We ground this in mechanisms that already run in Praxis, each with a diagnostic that reads it. The harmonic head (Section 3) builds a two-dimensional standing wave over (position, feature) from a learned grid of frequency amplitudes with irrational, frozen phases. A *static* amplitude grid produces the same field for every input: it is the corpus-average rhythm, maximum bias, a flat-by-construction prior. Making those amplitudes depend on the input adds *structured* variance - deviation that still respects the learned harmonic constraints. The encoder side shows the same physics from the other direction (Section 4). A continuous-latent codec reaches near-perfect reconstruction by phase-locking its latent features to a dominant rhythm [44]; the byte-latent encoder this run uses reaches the rhythm without a reconstruction objective at all, by cutting the stream at boundaries the data supplies and letting the cadence of the patch stream carry it. Two different bottlenecks, the same standing structure underneath. From these two we draw a scaling conjecture (Section 5): the mean costs linearly, interference costs logarithmically.

This paper therefore contributes: (1) a reframing of bias and variance as orthogonal axes of a harmonic latent space, grounded in a running implementation and its measured diagnostics; (2) a conjecture that consensus modeling scales linearly while harmonic modeling scales logarithmically, stated honestly as a conjecture; (3) the Praxis framework for testing it; and (4) a candid account of what is measured, what is conjectured, and what remains philosophy.

2 Definitions

The paper leans on a small vocabulary, used precisely. It is ordinary signal processing; the only commitment is to read the model's representations through it.

Harmonic. A representation is *harmonic* when it is carried by a superposition of oscillatory modes over a fixed frequency basis - amplitudes and phases on a frequency lattice, inverse-transformed into the signal - rather than by values in the raw signal space. The field $b = \mathrm{IRFFT}(a\, e^{i\varphi}/f^{\alpha})$ of Section 3 is the concrete instance: a is a spectrum of amplitudes, φ a set of phases, f the frequency lattice. The word is not shorthand for "oscillating" or "smooth"; it is the literal claim that the representation is a sum of fixed modes and that only their amplitudes vary.

Amplitude, frequency, phase, velocity. A mode at frequency f contributes $a_f \cos(2\pi f x + \varphi_f)$: *amplitude* a_f is how much of the mode is present, *phase* φ_f is where in its cycle it sits, *frequency* f is which mode it is. A mode's frequency is also its angular *velocity* - the rate its phase advances - so for a single mode "frequency" and "velocity" name the same quantity, and a velocity that varies across the signal is simply a frequency that changes (a chirp). In Praxis the amplitudes are learned, the frequencies are a fixed lattice, and the phases are *frozen*.

Frozen phases. The phases are seeded once - by Weyl equidistribution (Section 3) - and never trained. This is what fixes the basis: the same modes describe every input, and learning moves only the amplitudes. A representation whose phases are learned per input has no fixed basis and is not harmonic in our sense.

Geometry. The superposition of modes at one moment is a point - a *geometry* - in the latent space; inputs that share an amplitude configuration form a cluster of such points (the crystal-head centers, Figure 8). "Overlaid geometries" are several modes present at once. Distinct modes are orthogonal (the spectral theorem), so independent quantities can occupy separate modes without interacting - the property Section 3 uses to pull bias and variance apart.

Symbol. A *symbol* is a discrete, reusable geometry - a configuration the model commits to and recombines, not a one-off point. The crystal-head centers (Figure 8) are symbols in this sense: a finite, sparsely populated set of geometric units that recur across inputs. A symbol can be a geometry, but not every geometry is a symbol - the added content is that it is discrete and reused. We mean this in the literal, measurable sense (a clustered, low-dimensional set of recurring configurations), not the semiotic one. Read across training, this set is what grokking discovers - the periodic algorithm a network recovers only after its delayed memorize-then-generalize transition [39, 43]; a model with frozen phases is handed that basis at init, so it begins *post-grok* rather than working toward the transition. Reading computation as a *change in symbolism over time*

- which symbol is active, and how it shifts across position and depth - is the temporal companion to the static picture above: the symbol set is fixed, and what moves is which symbol is active, re-fitted per input (the *always grokking* conjecture, collected with the others in the Discussion).

Bias and variance, in these terms. *Bias* is the time-invariant part of the spectrum, the amplitudes shared across all inputs (the corpus rhythm); *variance* is the input-conditional deviation on top, carried by different spectral components and different parameters. Both are decompositions of the model's function over the *input* distribution - what stays fixed across inputs, and what moves with them - and not over resamples of the training set, which is the axis the classical decomposition's variance term measures. Two distinct quantities share the name; this paper means the first one throughout, and Section 3 says why that is the one an architecture can put coordinates on. The two do not stop being a spectrum: at any single decision boundary, a neuron's value still sets a *ratio* - how much corpus rhythm and how much input-conditional deviation pass downstream - a point between the ends. What the harmonic latent removes is the assumption that *one* such ratio stands for the whole model. There are many - a consensus of independent decision boundaries, each free to sit at its own point - and it is the aggregate of those ratios, not any single one, that lets bias and variance move as orthogonal coordinates rather than the two ends of a single dial.

Harmonics over time (Koopman). When the representation *evolves* - across sequence position, or across recurrent-depth steps - the harmonic claim is that it evolves approximately *linearly in the mode basis*: each mode carries its own frequency forward and the nonlinear dynamics become a linear operator on the spectrum. This is the Koopman view [31] of a dynamical system, and the sense in which we say the model works "in harmonic space." The claim has a boundary, and is therefore falsifiable: a representation whose basis depends on the input, that is not a superposition of a fixed set of modes, or whose dynamics are not approximately linear in any fixed basis, is *not* harmonic here. The universality - that *any* dynamics admits such a description - holds only in an infinite-dimensional limit; the working claim is that a *finite* harmonic basis approximates this model's behavior well enough to measure, and how well is an empirical question, not a definition.

Ghostmax. Attention normalizes scores with *ghostmax* (softmax1, [38]): the denominator is $1 + \sum_j e^{z_j}$ rather than $\sum_j e^{z_j}$, equivalently a single positionless key prepended to every row whose value vector is zero. The extra $e^0 = 1$ is an escape valve - when nothing in the context matches, the head spends its probability mass on the ghost and contributes nothing downstream, instead of being forced to average over irrelevant tokens. It is the dampening valve that lets a head go quiet rather than inject noise the harmonic field would then have to cancel. The boundary: a normalization that must sum to one over the real tokens has no quiet state, and is not ghostmax here.

Dropoff. *Dropoff* is the matched dual of ghostmax. Where the ghost is a zero sink at the *start* of a row that *absorbs* error, dropoff is a zero sink at the *tip* - the most recent position - that *injects* it. In its feature-wise form the value at the causal tip is scaled by an envelope $1 - e^{-(T-1-t)\,r_d}$ that is zero at the last position and recovers backward at a per-feature rate r_d, so the model cannot lean on the newest token and is pressed to carry its prediction in earlier structure. "Backward" here is the recurrent-depth axis, not sequence time: a causal pass never flows future to past. The boundary, and the present status: ghostmax is in production, but dropoff is a standing hypothesis - wired and runnable, not yet a measured result - and is dropoff only if mass demonstrably migrates off the tip.

3 The Mean and the Manifold

3.1 The tradeoff that was supposed to be a line

The bias-variance decomposition treats generalization error as the sum of a term that grows when the model is too rigid to fit the signal (bias) and a term that grows when it is too flexible and fits the noise (variance). The received wisdom is that you cannot reduce both at once with a fixed budget: regularize harder and bias rises; add capacity and variance rises. The tradeoff is drawn as a U-shaped curve with a single minimum, and model selection is the act of finding the bottom of that U.

That picture is correct when the model has one knob mediating both terms - one scalar of effective complexity. It stops being correct when bias and variance are produced by *different parts* of the architecture that can be optimized independently. A harmonic latent space is exactly such an architecture (Figure 1).

3.2 The harmonic field: a measured instance

The harmonic head modulates a model's hidden states multiplicatively by a learned field b - the instance this section measures, and a module a run either builds or does not: The current run builds neither, so this section describes the mechanism rather than the run.

$$h \;\mapsto\; h\,\big(1 + b\big), \qquad b[t,d] \;=\; \mathrm{IRFFT2}\Big(a[f_t, f_d]\, e^{\,i\varphi[f_t,f_d]} \,/\, f^{\alpha}\Big),$$

where t indexes position, d indexes feature, and $a[f_t, f_d]$ is a learnable grid of amplitudes over a two-dimensional frequency lattice. The phases φ are *not* learned. They are seeded by Weyl's equidistribution theorem [60]: cell (f_t, f_d) receives phase $2\pi\,\mathrm{frac}(f_t\pi + f_d e)$, which is equidistributed on the torus because $\{1, \pi, e\}$ are linearly independent over $\mathbb{Q}$. A radial $1/f^{\alpha}$ envelope imposes a pink-noise prior - which is not a free choice, and Section 5 prices it: the envelope attenuates exactly the high-frequency cells that interference capacity is counted in, so the prior costs capacity unless the amplitudes are free to grow against it.

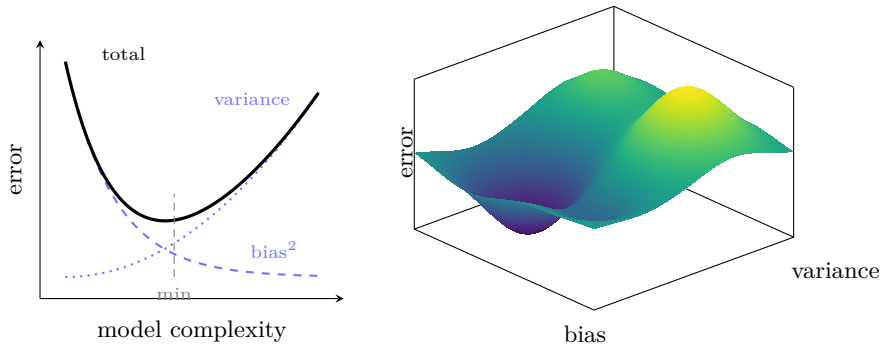


Figure 1: Schematic, not measured. **Left:** the received picture - one knob of effective complexity, so bias^2 falls while variance rises and total error traces a single U with one minimum; the best you can do is find the bottom of that curve. **Right:** when bias and variance are separate coordinates, error becomes a surface over a two-dimensional space - a landscape of basins and ridges navigated in two directions at once, not a single curve to slide along. The decoupling is what turns the line into the landscape.

Multiplicative coupling is deliberate: the upstream network cannot cancel the field by emitting $h(x) - b$, because the field scales features rather than adding to them, so learning is forced to flow *through* the field rather than around it.

Now read this through the bias-variance lens.

Bias is the static spectrum. With a fixed amplitude grid, $b[t, d]$ is identical for every input. It is a single rhythm imposed on all of the corpus at once - the population average. In the limit where the grid concentrates onto one cell, every feature oscillates to the same beat and the spectrum heatmap is flat: maximum bias, a model that has collapsed to the mean rhythm of its data. This is *mode collapse* in the literal sense the word already carries here - the single-mode limit of a harmonic spectrum, where the term for a distribution's mode and the term for a frequency mode name the same event - and it is measurable rather than diagnosed by eye. The head exposes this directly, reporting the Hoyer sparsity of the grid (1 = a single cell, 0 = uniform) alongside a live spectrum snapshot showing where the energy sits. The same reading survives when the field is carried not by a dedicated amplitude grid but by the periodic activation of a lighter transform - the per-feature frequency spectrum of a Serpent nonlinearity admits the identical Hoyer measure, so the concentration is a property of the harmonic representation rather than of one module that implements it. Early in training the heatmap is indistinguishable from noise; under a smoothness prior it develops a clean low-frequency band - the corpus rhythm crystallizing, with the Hoyer measure climbing as it does. (Figure 2)

One qualification on how to read that number, because it does not generalise across the two axes. A climbing Hoyer measure is the *static* grid doing its

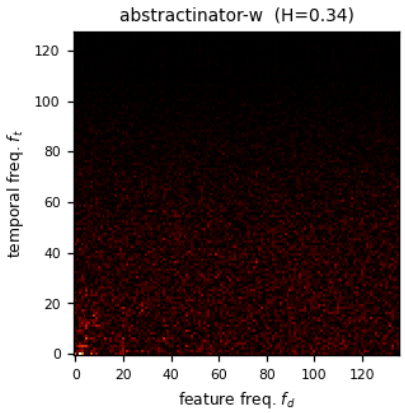


Figure 2: The static amplitude spectrum $|a[f_t, f_d]|$ for abstractinator-w - the learned grid that, inverse-transformed through the frozen Weyl phases, becomes the bias field. This is the heatmap the text describes: energy that settles into a low-frequency band (small f_t, f_d) is the corpus rhythm crystallizing, while a grid that stays uniform is a field still indistinguishable from noise. Hoyer concentration $H = 0.34$ quantifies it on the same $[0, 1]$ scale the prose names (1 = a single cell, 0 = uniform), computed from this very tensor - the same value the dashboard logs live. The pink $1/f^{\alpha}$ prior lives in the frozen basis, not in these amplitudes, so the grid shown is exactly what the head exposes.

job: concentration is what a population-average rhythm should look like. It is the wrong thing to want from the input-conditional delta introduced below, and Section 5 makes the reason exact - the configurations a harmonic field can distinguish are counted in the spread of its spectrum, so a delta that concentrates alongside the baseline has no room left to carry them. Concentration on the baseline and spread on the delta are not competing readings of one diagnostic; they are the two axes asking opposite things of the same instrument, which is the clearest operational form the decoupling claim takes.

Variance is the input-conditional deviation. The decisive architectural step was to stop forcing one static field on a corpus that mixes registers - dialogue, code, prose. A static grid must compromise across all of them, and the compromise degenerates into a content-blind side channel that the rest of the model has to route around. The fix makes the amplitudes a baseline plus an input-conditional delta:

$$a \;=\; a_{\text{static}} \;+\; \Delta_{\phi}(\text{context}),$$

where Δ_{ϕ} is a small network reading multi-scale pooled context, in the spirit of delay embeddings [51], with its output zero-initialized so the field begins as the trained static prior and *anneals* into input dependence. The static baseline carries the bias; the delta carries the variance; an L_2 penalty on the delta keeps

the baseline dominant. A diagnostic, the *delta norm* $\mathrm{rms}(\Delta)/\mathrm{rms}(a_{\mathrm{static}})$, reads how much variance the model is choosing to add. Zero means pure bias; a small bounded value means structured variance on top of a stable prior.

The delta's *magnitude* is not the only thing that has to be right, and the second requirement is easy to miss. Section 5 counts configurations by the spread of a spectrum, so the variance axis needs degrees of freedom, not just amplitude: a delta held to a handful of coefficients has nothing to spend however large the grid it modulates. We therefore size the delta's basis from the grid rather than by hand - a complete sine basis on each frequency axis, so the envelope can reweight the spectrum along features as well as along position, with the coefficients on the feature axis zero-initialized so the field is unchanged at step zero. The bound stays where it belongs, on the tanh depth and the L_2 penalty, which are soft controls that yield to data; a truncated basis is a hard cap that cannot. Two diagnostics say whether the room is used: the fraction of envelope coefficient energy on the feature axis, which is exactly zero at initialization, and the effective number of active modes, which is one. If both stay at their initial values the completion bought nothing and the delta was never the constraint - which is the outcome that would refute this paragraph.

The point is that these are *separate parameters with separate gradients*. The smoothness prior controls the bias term; the delta network and its regularizer control the variance term. Lowering one does not mechanically raise the other. The U-shaped curve has become a two-dimensional manifold, and we have coordinates on it.

One consequence is a stance on weight decay we state as a deliberate departure rather than a settled result. Weight decay is the canonical *single* complexity dial - a uniform shrink toward zero applied across every parameter at once - and its usual justification is the overparameterized regime, where capacity is abundant and pulling weights flat is how one buys a simpler, flatter solution. But a uniform shrink is precisely the one-knob instrument this section has been arguing against: it does not distinguish the static spectrum that carries the corpus rhythm from the input-conditional deviation on top, so it presses on bias and variance together, the coupling we built targeted controls (the smoothness prior, the delta regularizer) to avoid. In the underparameterized regime the objection sharpens, because there the learned geometry is not surplus capacity to be regularized away but the signal itself - the standing spectrum *is* the model - and a persistent pull toward zero, however slow, erodes exactly the structure the architecture exists to accumulate. We therefore largely eliminate weight decay on the geometry-bearing matrices rather than tune it, and we can do so because norm control here is structural rather than imposed: the interior update is orthogonalized (a bounded step), the harmonic latents are RMS-normalized, and the harmonic bases are fixed, so the growth that weight decay ordinarily guards against is already bounded by construction. This is a hot take, and we mark its falsifier plainly: the claim fails if the geometry-bearing weight norms grow without bound once the decay is removed, at which point a whisper of

decay - not the conventional amount - is the correct repair. What the argument does *not* touch is the targeted regularizers themselves; the delta penalty that holds variance in check is a control on one axis, not a global dial, and it stays.

3.3 Temperature as a coordinate, not as noise

Sampling temperature is usually described as noise injection. On the bias-variance manifold it is better described as a position dial. At low temperature the model commits to the learned rhythm - the high-bias corner. As temperature rises it explores within the basin the harmonic constraints define - structured variance. Past a critical point it begins to violate those constraints, which reads as either failure or discovery depending on what you were hoping for.

3.4 Which variance, and where it is spent

One disambiguation, because two quantities share the name and the argument above only works for one of them. The classical bias-variance decomposition measures variance across *resamples of the training set*: hold the input fixed, retrain, and ask how much the prediction moves. That is a property of the training procedure, and an architecture has no direct coordinate on it. The quantity this section has been decomposing is different - it is the variation of the learned function across the *input* distribution, the part of the field that moves when the context moves. Decomposing a function into a constant term plus input-conditional terms, and attributing variance to each, is the functional-ANOVA construction that sensitivity analysis has used for decades [47]; the static amplitude grid is the constant term and $\Delta_\phi(\text{context})$ is the first-order term. Reading the section under the other definition inverts it, because there a model that responds more to its input has *lower* bias rather than more variance.

Naming the axis this way also says why the harmonic basis is the natural place to hold it. A power spectrum is the transform of an autocovariance, so a spectrum is already a variance map; what distinguishes the two ends of our axis is not amplitude but the *kind* of spectral mass. A pure line spectrum - discrete atoms at fixed frequencies - is deterministic and repeats forever, which is precisely the corpus rhythm. Spread that is not concentrated into lines is the continuous part, and it is what carries input-conditional structure. Koopman theory makes the same cut for dynamics, where the point spectrum holds the eigenfunctions that recur and the continuous spectrum holds the mixing part that has no eigenfunction at all [37, 31]. Our two axes are that decomposition read on a language model: bias is the atomic part, variance the continuum.

This sharpens a diagnostic the paper already leans on. Hoyer concentration measures how tightly mass sits in a few cells, which tracks the atomic reading well and says nothing about how the remaining mass is distributed. The measure the argument actually wants is the *ratio* of line mass to continuum mass, taken separately on the baseline and on the delta - and the prediction the decoupling claim makes is that the two should move in opposite directions. If they move

together, the axes are not separate parameters after all, whatever the gradient graph says.

There is a second, harder question the framing raises and this paper does not settle. Representing a distribution and *using* one are different things, and a readout that summarises a distribution by its expectation discards the spread by construction, however carefully the spread was built. A mixture over positions consumed as a weighted average of values is a mean; so is a soft assignment to a set of centers. The distribution shapes *which* mean, and then the mean is what travels downstream. Whether a model gains anything by carrying the spread instead of collapsing it is an empirical question with at least one clear affirmative answer elsewhere: in reinforcement learning, replacing the expected return with the full return distribution improved the resulting policy even though that policy still acts on the mean [6]. The representation was worth more than the summary. We record the analogous question here without claiming the analogous result - it is a direction, and its falsifier is the ordinary one, that a model given room to carry spread declines to use it.

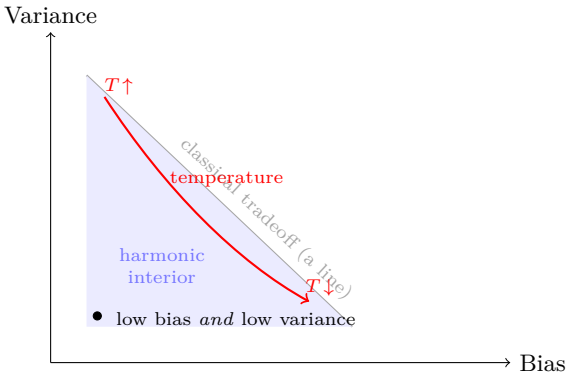


Figure 3: Bias and variance as orthogonal coordinates. Classically a model must sit somewhere on the gray line: less of one means more of the other. When a static spectrum supplies bias and an input-conditional modulation supplies variance, the two decouple, and the interior (shaded) becomes reachable - lower bias *and* lower variance than any point on the line. Temperature is a path through this space, not noise added at a fixed point.

4 Harmonic Latent Space

4.1 Entropy patching and the corpus beat

This run reads raw bytes through a byte-latent encoder [41]: the stream is segmented into variable-length patches at boundaries the data itself supplies - whitespace, in the configuration we run, so a patch is roughly a word - and each patch is compressed through a bottleneck to a single vector the global

transformer then operates on. There is no reconstructing codec here and no per-patch latent to phase-lock, but the same geometry appears one level up. Because the boundaries follow the content rather than a fixed stride, the patch stream already carries the corpus's own cadence: patches vary in length exactly where the text does, and the sequence of patch vectors is a rhythm read off the data instead of imposed on it. The harmonic head then writes its standing wave over those patch features, and the bias/variance reading is unchanged - a static spectrum is the average patch rhythm, an input-conditional modulation is the structured deviation. Where a continuous-latent codec phase-locks its latent to the beat, the byte-latent encoder lets the beat fall out of where it chose to cut.

4.2 The outer representation

There is a third way to read what this architecture trains for, orthogonal to the bias-variance manifold and to the Koopman dynamics: not *how* it represents, but *where in the sequence the learning pressure falls.* A standard causal model carries a next-token objective at every position; it learns the running interior, the representation as it unfolds one step at a time. A masked model hides interior positions and learns to fill them in; it learns the inner representation by infilling. Both place their pressure inside the window. This architecture places it at the *ends.*

The same two temporal poles are present here, but they are carried by the architecture rather than written into the loss. This run trains end-to-end on byte cross-entropy alone - there is no second reconstruction objective over the patch, and no per-patch latent trained to round-trip. What creates the pressure at the ends is the bottleneck's position in the stack. The local encoder must compress a span of bytes into a single patch vector well enough that the local decoder can restore those bytes, which anchors the *already-consumed* span; the global transformer over patch vectors must carry that representation forward to the next patch, which anchors the *unobserved* continuation. One objective, two poles: the byte loss lands at both ends of the bottleneck and passes through its interior without ever making that interior a target. That is where the pressure is designed to fall, which is not the same as where it lands.

The patch boundaries themselves are content-adaptive, so the model is not imposing a grid on the stream but reading one off it - the cut points follow the data's own segmentation rather than a fixed stride, and the patch stream inherits that cadence. Where a reconstructing codec pins its ends with an explicit loss at each, a byte-latent encoder pins them with a bottleneck the gradient has to pass through in both directions. The reading in the rest of this section - that learning pressure migrates to the rim of the window rather than its middle - is unchanged; only the mechanism that puts it there differs.

This has a precise literary shape. A fixed alphabet and a fixed page length already contain every text that could be written - Borges' Library of Babel, whose modern realization stores nothing and instead returns any page from a

reversible address-to-content function [4]. The harmonic patch is such a library: the frozen phases and the fixed lattice fix the alphabet and the page, and the amplitudes are the address. Reconstructing the consumed patch and predicting the next are then not searches through that space but its two neighboring pages, read off by the same invertible transform - the linear solve that carries the field between coefficients and content, worked at the edges of a fixed-length window rather than its interior.

Carry that reading forward and the genre of the problem shifts. If a patch is an address into a fixed continuous space, then underneath the tokens the interior is *time-series modeling over continuous samples* - statistical fitting of a real-valued signal against a fixed basis, the discrete vocabulary only a boundary condition imposed at the edges and the interior continuous throughout. The codec's continuous latent is the native object and the tokens the incidental one; the architecture's already-validated instances are literally continuous series, not text.

Stated that loosely the reading buys nothing. Any architecture that embeds discrete symbols in $\mathbb{R}^d$ can be narrated as fitting a real-valued signal, and a description that fits every architecture distinguishes none of them. It is worth keeping only where it predicts something the token account does not, and there are two such places.

The first is *where the error lives*. A fitted signal that mostly holds its standing shape needs correcting only where it departs from it, so the computation is *event-driven* and *sparsely activated* by construction: most positions are null work and compute falls on the deviations. That is the regime the conjecture below pushes to its limit, and the concentration of the reported delta norm already tests it.

The second is *off the sampled interval*. A fitted signal has values where no sample was taken; a sequence of discrete tokens does not. If a state is genuinely an address into a fixed basis rather than a summary of the symbols it consumed, then reading it at positions it has not consumed should degrade *gradually* with distance, and should not break at the patch boundary - the boundary is an artifact of how the input was cut, not a feature of the underlying signal. The falsifier is sharp: a read that is accurate inside the consumed span and collapses at its edge is a model that memorized a window, and the address language should be dropped.

The two attention sinks of Section 2 mirror the two objectives in the same geometry. Ghostmax is a zero sink at the head of every attention row - positionless, always reachable, the place a query falls back to when the past holds nothing it needs. Dropoff, when enabled, is a zero sink at the tip - positioned on the most recent token, the place that forces a query off the easiest signal so it cannot simply copy the present. A sink that absorbs at the start and a sink that injects at the end: together they pull the model toward the boundaries of what it can see and away from the comfortable middle. We call the result the *outer*

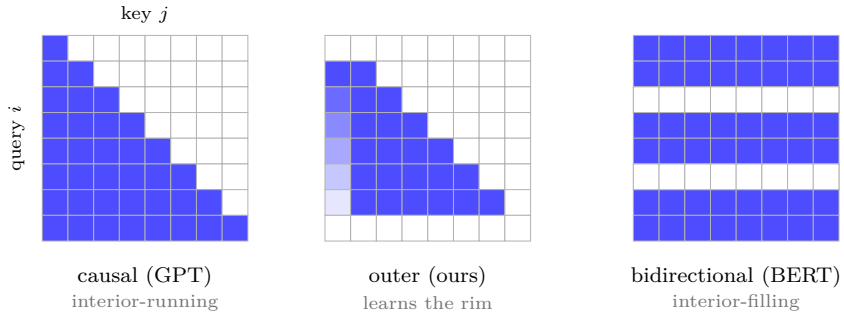


Figure 4: Where the learning pressure falls, read as an attention mask (rows = query position i, columns = key position j; a blue cell means i attends to j). **Causal (GPT)** is lower-triangular: every position predicts the next from its past, so pressure runs along the interior. **Bidirectional (BERT)** is the full square with a few interior query rows held out as masked targets (the white bands): pressure fills the interior. **Outer (ours)** keeps the causal triangle but drops the first and last query rows to sinks - ghostmax at the head, dropoff at the tail - so the objective is anchored at the edges of the window rather than its middle: it learns the rim.

representation - the model is incentivized at the head and the tail, the edges of the window, rather than along its interior. It is the complement of both standard regimes (Figure 4): causal learning is interior-running, masked learning is interior-filling, and this learns the rim.

The rim reading earns a term of its own. The received picture of causal language modeling - the figure this field has redrawn a hundred times, tokens in boxes, "The quick brown fox jumps" rendered left to right, a little arrow hopping from each box to the next - encodes an assumption that is rarely stated: uniform *information density*. Every token is a self-contained semantic unit, equally represented, to the best of the mean's ability, and the objective enforces the assumption by landing the same loss at every position. Call information density the share of the representation's distinguishing content carried at each position - and, once a depth loop exists, at each step of that loop. In the boxes-and-arrows regime it is a per-token constant by assumption. The harmonic model breaks it by construction (Figure 5): the lowest total-frequency modes span the whole window and move slowest, so the head of the sequence carries a slow, echoing hum - the corpus rhythm, the bias axis, the sink that absorbs - while the input-conditional deviation rides the leading edge as a chirp in otherwise settled bias - the variance axis, the sink that injects. Hidden states are extremely stable at the head and unstable at the tip, and the linear token-to-token relationship of the received picture is gone: density stops being a constant and becomes a shape.

The strong claim - that a consensus model *cannot* do this - would be false, and the way it is false is the best evidence for the weak one. Vanilla transformers

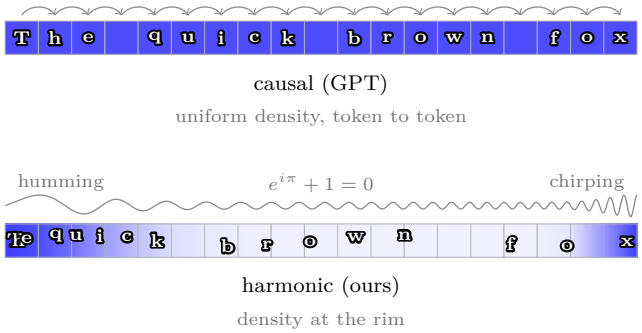


Figure 5: The received picture and the harmonic one, over the same sequence (position runs left to right; shade is *information density*, the share of the representation's distinguishing content carried at each position). **Top:** the figure every paper redraws - tokens in boxes, an arrow from each to the next. Its unstated assumption is uniform density: every token a self-contained semantic unit, equally represented, to the best of the mean's ability. **Bottom:** the harmonic model pushes density to the extremes of the window - a slow, echoing hum at the head, where the low-frequency modes that span the whole context anchor the corpus rhythm (the sink that absorbs), and a chirp at the tip, where the input-conditional deviation rides the leading edge (the sink that injects). The same characters, freed of their boxes, pool toward the head, cross the cell walls, and ride the field's slow undulation - the interior goes quiet, and the linear token-to-token relationship is broken by construction. Where the top panel hops token to token, the bottom draws the field itself: one continuous wave, humming at the head, chirping at the tip, its amplitude the density envelope - and Euler's identity between the two gerunds for what the wave never stops doing: looping. Schematic, not measured - the conjectured profile this section names, ahead of the metric that will read it.

break the uniform picture on their own: attention mass collapses onto the first token, and a handful of massive activations anchor it there, in essentially every large causal model measured [61, 50]. Gradient descent visibly strains to build a hum at the head of an architecture that gives it no coordinates for one, and the sink it improvises is a bug the architecture tolerates rather than a feature it provides - the gradient cannot tell bias from variance inside a single scalar loss (Section 3), so it detects the significance of the split and overshoots it. The harmonic model makes the same structure constitutive. Ghostmax is the head sink by construction, dropoff is the tip sink by construction, and the frozen phases fix which modes are slow and which are fast, so hum-at-the-head, chirp-at-the-tip has an address before training begins. Encode the algorithm into the architecture and the model reaches solutions the gradient would otherwise miss - or find only as this unstable, undirected approximation.

For now, information density is a term and a conjecture, not a metric; the quantitative profile - deviation read per position, per depth step - was deferred until it could be reported rather than promised. But the reading already names

its refutation, and it is the sharpest one this section offers. If density is pushed to the extremes, hidden-state deviation must grow with recurrent depth under a monotone positional gradient: early positions settle toward a fixed point while the tip stays live, and the profile should steepen as the loop deepens - and the same shape should reappear wherever a recurrence is harmonized, not only in the depth loop here. A per-position deviation profile measured flat, or one that fails to steepen across depth, kills the reading outright. The per-position halting distribution, where the adaptive-depth loop runs, already reads this profile through compute - pressure that halts early positions early and holds the tip open - so the conjecture arrives with its instrument already logged, waiting to be read.

That falsifier, as first stated, has a blind spot, and closing it sharpens the term. A deviation profile reads magnitude, and magnitude can miss the event: at a decision boundary a single bit's worth of change - one neuron, one activation value - can move the hidden state from one basin of the latent geometry to another while remaining silent in norm. Density asks how much is carried at a position; *geometry* asks which shape is expressed there, and the second is the coordinate system the architecture actually computes in. So the profile must be read twice: in norm, and in *symbol occupancy* - which cluster of the learned codebook the state expresses (the centers of Figure 8), per position, per depth step. In those coordinates the prediction reads: the hop rate between symbols rises from head to tip and steepens as the loop deepens. A silent flip that changes membership registers at full weight regardless of magnitude, and the conjecture is now falsifiable in both coordinate systems rather than escapable through either - flat in norm *and* flat in occupancy leaves it nothing to hide behind.

Where would such a flip come from?

We do not have an answer for this run, and the honest move is to say so rather than borrow one. The forced-error reading we develop elsewhere rests on gradient clipping that binds at nearly every step, which is a property of the continuous-latent regime and not of the architecture: the threshold is the same here, but the gradient norm rarely reaches it, so descent keeps its magnitudes and nothing is systematically truncated. Nor does this run carry a quantizer, whose straight-through estimator would supply the same blindness by construction. Both of those are variation the gradient cannot see; this configuration has neither, and we will not manufacture a third.

That leaves the question of Section 2's silent flip open here, which is worth stating plainly because it makes this configuration the control. If symbol occupancy still hops - if the state moves between clusters of the learned geometry without a forced error anywhere in the update rule - then the flips were never the clipping's doing, and the reading that ties them to it is weaker than it appears. The occupancy profile is measured the same way in every configuration, so the comparison is available whenever the runs are put side by side.

Wherever such a source is present, it forces one refinement on Section 3: the orthogonality of bias and variance is an instantaneous claim about the loss surface, and it can hold pointwise while the axes still feed each other across training - the variance mapping each flip's consequences, the bias consolidating the map, the consolidated spectrum moving where the next flip lands. The axes are orthogonal in space and coupled in time, and nothing in the decoupling argument requires otherwise.

Pushed one step further - and now firmly into conjecture - the same machinery suggests a regime in which the model *mostly does not transform at all*. A head that falls silent through its ghost is an identity on the residual stream, contributing nothing; and with the input-conditional delta held small against the standing static baseline (Section 3), the field a sequence sees is dominated by the input-*independent* corpus rhythm. The transform is then nearly the same for every input, and what separates one sequence from another is a sparse scatter of large deviations on a shared standing shape - the smoothness prior carrying that shape forward, each feature entering with a velocity and only now and then kicked off it. Read this way the computation is *event-driven* rather than a fresh dense map at every step, closer in spirit to the sparse activity of spiking systems - a point of contact with biological plausibility we flag as motivation, not as a claim that the architecture is neural. It is checkable in diagnostics already reported: the delta norm measures how much deviation is added, and how *concentrated* that deviation is across positions and features is the sparsity this reading predicts - a diffuse delta would refute it. It leaves open the further question, which we do not attempt here, of whether a downstream consensus - a vote taken over decoded candidates rather than a single argmax - could itself raise such events and fold them back into the field.

This is the least settled part of the picture, and we mark the seam honestly. The reconstruction-and-prediction duality is in production and measurable - reconstruction fidelity and the conditional gap between marginal and conditional next-latent prediction are both reported. The outer-representation reading, and the dropoff sink that would complete its symmetry, are a conjecture: the claim that learning pressure migrates to earlier positions is exactly what a measurement of attention mass would confirm or refute. We offer the axis as a third coordinate to look along, not yet as one we have surveyed.

5 The Scaling Conjecture

5.1 The watch

Hold the title literally for a moment. A watch is a mechanism of intricate, meshed gears, and what makes it a watch is that the gears turn *in unison*. Drive one and you drive them all, because the teeth couple them into a single body. Now try to run the watch by turning each gear in sequence - one at a time, the rest held still. You cannot. The logic does not merely become inefficient;

it stops being a watch. The architecture, the mesh itself, forbids sequential operation. The coupling is the whole point.

Autoregressive consensus modeling is the attempt to turn the gears in sequence. It emits one position, conditions on it to emit the next, and accumulates the answer a tooth at a time. For a coupled system this is the wrong shape of computation, and its cost is the cost of enumeration: linear in the parts.

The harmonic field turns the gears in unison. The frozen Weyl phases are the teeth - the coupling, cut once and never re-cut, that ties every feature to every position in fixed relation. The amplitude grid is the single mainspring. One inverse transform releases it, and the entire field, every (t, d) cell at once, assumes its shape in a single coupled motion rather than a loop over positions. That is the mechanical content of "logarithmic": the whole mechanism moves together, so the cost tracks the number of independent rhythms it carries, not the number of positions it must cover.

And the field is *rebuilt*, not advanced. With input-conditional amplitudes it is re-derived from the input rather than carried forward and nudged; in the limit, the entire shape is reconstructed at every decision boundary, all of them, the whole computation re-formed around the present input. What the model learns under the smoothness prior - which asks the field at one position to predict the field at the next - is not a sequence of outputs but *the ability to hold a shape over time*: to keep the gears meshed, to keep the watch a watch, while the hands sweep. The frozen phases protect the coupling; the smoothness prior protects the form.

This is why the geometric framing is not ornament. Nearly anything we can imagine can be written as a shape - a configuration in some space, a standing wave, a mesh of constraints that must move together or not at all. A model that builds shapes and protects them over time models that structure directly, in unison, where a model that enumerates is forever turning one gear and hoping the rest follow. There is something here, and the rest of this section is an attempt to say how much.

This is also why the bet is made small. Praxis is built and tested on the order of a million parameters - far below the scale where a model can simply store the corpus - because the claim is only meaningful where there are not enough parameters to memorize. At that size capacity cannot come from count, so it has to come from somewhere else, and the architecture's wager is that it comes from computation: the recurrent depth that turns the same small mesh again and again, and the harmonic coupling that moves every feature in one motion. Capacity is paid in the turning of the gears, not in their number.

The forced structure is the mechanism by which that payment buys anything. A harmonic field, a continuous latent, an autoregressive factorization - each is an inductive bias that makes the model compute the way a particular mathematics computes, importing that domain's structure rather than treating the architecture as neutral ground. Some of what is imported is exact: an autore-

gressive factorization is a proper normalized likelihood by the chain rule, not an approximation of one. Most of it is structure, not a theorem - a smoothness, a basis, a fixed point to iterate toward - and we keep the distinction honest. The stronger reading, that computing in a domain's mathematics confers that domain's guarantees, is the question the framework exists to test, not a result it assumes; the conjecture above is its measurable edge. If interference scales as $\log N$, a model with few parameters but many coupled turns should reach what a far larger consensus model reaches - which is exactly the experiment a million-parameter run is for.

5.2 Linear consensus, logarithmic interference

Here is the claim the rest of the paper has been building toward, stated as a conjecture because that is what it is.

A model that represents its corpus as a *mean* - a consensus, an average of examples - must spend resources roughly in proportion to the number of distinct patterns it needs to express. To hold N patterns it allocates capacity that grows with N. This is the linear regime, and it is consistent with why scaling laws ask for ever more parameters and tokens to buy each further increment of capability [28].

A model that represents its corpus as *interference* - a superposition of K frequency components - addresses a combinatorially larger space. K components with learnable amplitudes and frozen, equidistributed phases produce a field whose distinguishable configurations grow exponentially in K, because it is the *pattern of constructive and destructive interference* across components that carries information, not the components individually. To express N configurations you need on the order of $\log N$ components. This is the logarithmic regime. The same asymmetry shows up in time: the transform that builds the field is $O(n \log n)$, and a single transform realizes the whole interference pattern at once rather than enumerating its parts.

The exponential count is the half of that claim the rest of it rests on, so it is worth deriving rather than asserting. It follows from one property of the basis, and the property is less exotic than the machinery around it suggests.

Modes at distinct frequencies are orthogonal *whatever their phases*. The phase of a sinusoid does not affect its inner product with a sinusoid at a different frequency - the cross terms integrate over a full period either way. So the map from a coefficient vector to a field is a scaled isometry: distances in coefficient space are distances in field space, exactly, with no compression and nothing lost in the transform. Whatever the model can distinguish among its amplitudes, it can distinguish among its fields.

That is what makes the counting argument sound rather than merely suggestive. Fix a resolution - the finest difference in a coefficient the rest of the network can still act on. The distinguishable amplitude settings are then a grid in K

dimensions, and a grid in K dimensions has exponentially many points in K. The isometry carries every one of them to a distinct, equally separated field. Expressing N configurations therefore costs $\log N$ components, which is the conjecture's logarithmic side, now with a derivation under it.

This also settles what the frozen phases are for, and it is not orthogonality - orthogonality was free. Freezing buys a *fixed* basis: the same modes describe every input, so an amplitude configuration means the same thing from one forward pass to the next and the count above is a count of stable, reusable configurations rather than of momentary coincidences. Weyl equidistribution chooses *which* fixed basis, spreading the phases so no region of the torus is left unaddressed. Neither is doing the work the isometry does.

Part *(iii)* is a constraint the derivation produces on its own, and it cuts against a reading of the spectrum this paper otherwise encourages. The count is in *effective* coefficients - amplitude after the $1/f^{\alpha}$ envelope. If the raw amplitudes stay within a fixed range while the envelope attenuates the high frequencies, the capacity does not grow exponentially in K at all; for $\alpha > 1$ it converges to a constant, and adding components buys nothing. Interference capacity is available only if the model spends amplitude against the envelope where the envelope is suppressing it.

Read that alongside the concentration measure and the two axes come apart again, in a way that is checkable rather than interpretive. A spectrum collapsing onto a few low-frequency cells - rising Hoyer concentration - is the bias axis doing exactly what Section 3 says it should: the corpus rhythm crystallizing. But it is also, by the proposition, a spectrum of low interference capacity. The two readings are not in conflict, because they are readings of different parameters: the static grid is where concentration belongs, and the input-conditional delta is where capacity has to live. The falsifier is on the dashboard already. If the delta's spectrum concentrates alongside the static grid rather than staying spread - and in particular if its high-frequency mass decays with the envelope instead of resisting it - then the variance axis has no room to carry configurations, and the logarithmic regime is unreachable in this architecture whatever the counting argument permits.

Proposition 1. *Let the harmonic field over n positions be $b(c)[x] = \sum_{k=1}^{K} c_k \cos(2\pi f_k x/n + \varphi_k)$ with distinct integer frequencies $0 < f_k < n/2$ and any fixed phases φ_k. Then:* (i) *the modes are mutually orthogonal with squared norm $n/2$* whatever the phases are*, so $c \mapsto b(c)$ is injective and $\|b(c) - b(c')\|_2 = \sqrt{n/2}\,\|c - c'\|_2$;* (ii) *consequently, if each coefficient is resolvable to a precision ε over a range R, the field admits at least $(R/\varepsilon)^K$ mutually distinguishable configurations, so expressing N of them requires only $K = \lceil \log N / \log(R/\varepsilon) \rceil$ components;* (iii) *the count is in* effective *coefficients $c_k = w_k a_k$. Under a fixed envelope $w_k = f_k^{-\alpha}$ with raw amplitudes a_k confined to a fixed range, $\log(\mathit{count}) = \sum_k \log(1 + Rw_k/\varepsilon)$, which converges for $\alpha > 1$ and grows only like $\log K$ at $\alpha = 1$: capacity saturates in K unless the model grows high-frequency amplitude against the envelope.*

Proof. *(i)* Expand the inner product of two modes: $\sum_{x=0}^{n-1} \cos(2\pi fx/n+\varphi)\cos(2\pi gx/n+\psi) = \frac{1}{2}\sum_x [\cos(2\pi(f-g)x/n+\varphi-\psi)+\cos(2\pi(f+g)x/n+\varphi+\psi)]$. For $f \neq g$ with $f, g \in (0, n/2)$ neither $f-g$ nor $f+g$ is $\equiv 0 \pmod n$, so both sums are geometric series over a full period and vanish. For $f = g$ the first term contributes $n/2$ and the second vanishes because $2f \not\equiv 0 \pmod n$. The Gram matrix is therefore $(n/2)I$ regardless of φ; the map has full rank K, and Parseval gives the stated isometry.

(ii) An isometry carries an ε-separated set in coefficient space to a $\sqrt{n/2}\,\varepsilon$-separated set of fields. The axis-aligned grid of spacing ε on $[0, R]^K$ is such a set and has $(\lfloor R/\varepsilon \rfloor + 1)^K$ points. Taking logarithms and solving for K gives the stated bound.

(iii) Substitute $c_k = w_k a_k$ and bound $\log(1 + Rk^{-\alpha}/\varepsilon) \leq Rk^{-\alpha}/\varepsilon$; the sum converges by comparison with $\sum_k k^{-\alpha}$ for $\alpha > 1$ and diverges logarithmically at $\alpha = 1$.

The count is a statement about which fields are *representable and mutually resolvable*, not about which are reachable by gradient descent - the representational layer of the three-way split this paper draws elsewhere, and the layer on which universal approximation is likewise silent about cost. □

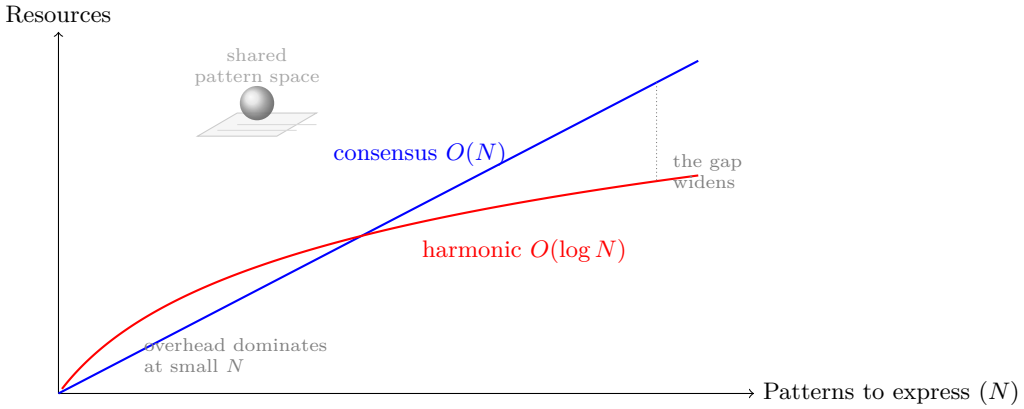


Figure 6: Conjectured scaling regimes. To express N patterns, consensus modeling - learning the mean - spends resources growing linearly in N (blue). Harmonic modeling composes patterns by interference, and $K \sim \log N$ components suffice, so its cost grows logarithmically (red). The harmonic basis carries fixed overhead, so it loses at small N and wins at scale, where the gap widens without bound. Both must ultimately represent the same target (the shared pattern space, top); the claim is about the cost of getting there, not the destination. This figure states a conjecture, and replaces the speculative measured-curve figure of earlier drafts.

5.3 The exponential edge

Why should a learned network be able to live in the interference regime at all? Because the transformer already computes on an exponential substrate. Every attention layer is a softmax, $\exp(\cdot)/\sum\exp(\cdot)$, which amplifies small differences and converts addition into multiplication; the network is kept numerically alive only by normalization holding activations in the narrow band where the exponential is both sensitive and stable. Euler's identity, $e^{i\pi} = -1$, ties that exponential directly to periodic structure, which is why a harmonic field seeded with π and e is not an arbitrary choice of decoration but a resonance with the machinery already present. The harmonic head writes interference patterns onto a substrate that was always exponential; it makes explicit a structure the softmax was implicitly using.

There is a sharper version of this available, and it is empirical rather than rhetorical. A network's arithmetic is not the arithmetic of the reals it is written in terms of; it is floating point, and floating point stops being linear near zero, where underflow and denormals bend addition and multiplication away from the operations they approximate. That bend is ordinarily treated as a defect to be normalized away. It is also a resource: a network built entirely from linear layers - no activation function anywhere - can be trained to exploit it, and doing so lifts the same architecture from the 92% MNIST test accuracy a linear model is entitled to, up to 96.7% [15]. The nonlinearity a deep network is supposed to require was supplied, in that experiment, by rounding error alone.

Two details of how it had to be done matter more here than the accuracy. The first is that the activations were deliberately held small enough to sit in the range where float32 is nonlinear - the result depends on parking the computation at a numeric boundary rather than at a comfortable scale, which is the same discipline normalization performs for the softmax, run toward the opposite edge. The second is that gradient descent could not find it. Symbolic differentiation differentiates the idealized operation, not the one the hardware executed, so backpropagation is structurally blind to a nonlinearity living in the low-order bits; the parameters had to be found by evolution strategies, which need only the loss and never the derivative. Variation the gradient cannot see, selection performed by the loss: the blind watchmaker of Section 8.5, arrived at from the numerics rather than from the argument.

We take two things from this. Numerically, that a representation living close to the edge of its own precision is doing computation there rather than merely tolerating error - so *measurement-induced rounding*, a cast or a snapshot that nudges a value across one of those boundaries, is a perturbation of the computation and not just of the report of it. And methodologically, that the interesting nonlinearity in a system can be the one its optimizer is constitutionally unable to look at.

5.4 Tractability is observer-relative

If interference lets a model represent exponentially many configurations with logarithmically many components, then a search that is combinatorial when enumerated becomes, in this representation, the reading-off of an interference pattern computed in a single transform. The hardness did not vanish. It moved into the choice of basis. And that relocation is the real claim, which we will state without hedging: in a continuous, nested representational space, P and NP are not two discrete bins a problem falls into once and for all. They are the ends of a spectrum, and where a given instance sits on that spectrum is set by the representation doing the solving - by the observer.

This is the same thesis the rest of the paper has been making, carried to its conclusion. If attention constructs which patterns manifest (Section 7), and if the basis a model occupies decides whether a structure costs linear or logarithmic resources to express (Figure 6), then "intractable" cannot be a property of the problem alone. It is a property of the pairing between a problem and a solver. The instance that is exponential to one architecture - one that can only enumerate, that holds no basis in which the instance is smooth - is approximable in near-constant work by another that happens to carry the right harmonic. Some observers are simply better at collapsing NP-shaped problems toward P; others are blocked, not by the problem, but by their own architecture, perspective, or alignment. The watchmaker who cannot perceive the gear cannot build the watch, however long he is given. The difficulty was never only in the watch.

We keep exactly one boundary, and we keep it because it sharpens the claim rather than softening it. This is not a statement about the formal worst-case question - whether the classes P and NP coincide over all inputs and all machines [10]. That question is deliberately indifferent to who is asking; it abstracts the observer away on purpose. The claim here is the complementary one, and abstracting the observer away is precisely what it refuses to do: for the bounded, structured, approximate problems that real minds and real models actually face, tractability is observer-relative, continuous, and movable - and moving it is what changing your representation *is*. The weak, measurable form is already in the framework: for a family of problems one can record the scaling slope each architecture buys, and watch a single instance migrate along the P-to-NP spectrum as the basis changes underneath it.

6 The Praxis Framework

Praxis [8] is a registry-based framework in which every component - encoder, decoder, attention, optimizer, loss, router, embedding, head, memory - is swappable, and new implementations become available through registration without touching core infrastructure. Read in light of the preceding sections, its components are not an arbitrary catalogue; each one shapes the representational geometry in a different way, and the framework's purpose is to let those geome-

tries be combined and compared rather than to scale any one of them.

- **ByteLatent compression** [41] forces re-attention through an entropy-based patching bottleneck, selecting which correlations survive aggressive dimensionality reduction.
- **MonoForward training** [34] detaches gradients between layers and trains each with a local objective, turning one global trajectory into many independent ones - whether run in-process, across a Ray actor per layer, or over the remote-expert pool, where a slow peer never blocks the rest.
- **Reversible residuals** [23] require invertibility throughout the forward pass, forbidding destructive compression and so forcing reconstruction-preserving representations.
- **Attention variants** (multi-head, sliding-window, gated, structured $O(n \cdot c)$, and others) each impose a different relational geometry on the sequence.
- **Mixture-of-experts routing** [45] and **tool integration** let several geometries coexist and let external, bias-free computation enter the loop.

6.1 How the model is trained

A paper that argues architecture is not neutral ground owes the reader the same disclosure about its optimizer, because an update rule is a choice of geometry too. Steepest descent is only defined relative to a norm, and which norm a rule descends under decides which directions in weight space are cheap. The framework therefore registers optimizers the way it registers heads and attention - as swappable geometry - and this section reports the ones the present run selected.

This run optimizes with **AdamW**. AdamW descends under a diagonal, per-coordinate rescaling of the gradient - each weight moves on its own axis, and the geometry the step respects is the one in which those axes are already the right ones. It is the field default, and this framework treats it as the control rather than the goal.

The configured settings are learning rate 0.001; betas 0.9, 0.95; weight decay 0.1.

Wrapped by Schedule-Free [12] (no learning-rate schedule; the wrapper Polyak-averages the iterate instead, so there is no decay curve to tune or to end).

Credit is assigned by back-propagation locally, with the remote-expert pool configured. Where peers are attached, their layer-wise updates are Mono-Forward by construction - no global backward pass to synchronize, so a slow peer lags rather than blocks and its update lands on a later step. With no peers attached the run is an ordinary single-process one, so this line reports what the configuration permits rather than how many peers happened to be online. This is one point in a space the framework leaves open: back-propagation is the default rather than the commitment, and Mono-Forward, the remote-expert swarm, forward-forward and energy-based objectives are registered alternatives

that change where credit comes from without changing the model they train. Which of them a run uses is a configuration line, and it is recorded here for the same reason the architecture is: so that a result can be attributed to something.

The configuration hierarchy (Environments > CLI > Experiments > Defaults) makes arbitrary combinations reproducible: different attention per expert, optimizers per layer, frequency offsets [30, 52] per head. 110 training runs are documented in the run history. This paper does not present a benchmark table, by design: the framework is the instrument, and the experiment that matters is the one the reader runs to test the scaling conjecture for themselves.

7 Attention as a Constructor

The framing that motivates all of this is older than the harmonic head. Across physics and neuroscience, observation appears to participate in what is observed rather than passively recording it: measurement affects the quantum system [25], attention modulates cortical responses to identical stimuli [57, 29], and predictive-processing accounts make perception an attention-weighted construction rather than a bottom-up readout [18, 9]. If attention constructs which patterns manifest rather than filtering pre-existing ones, then architectural diversity in attention is not a tuning detail; it determines what is learnable at all.

The measurement analogy invites a sharper question than the one it usually settles for: if observation participates in the observed, does *our* act of observing the model - snapshotting its spectrum, sampling its state - change what it learns? The quantum reading is metaphor, and we keep it as such. But a harmonic field is a standing wave over the full (position, feature) computation, and therefore non-local: a perturbation injected at one cell rides the field to the rest rather than staying put. So a harmonic model should be *more* sensitive to impure observation than one whose features are independent, and the weak, measurable form of the question follows directly. Three of its artifacts are mundane and real. First, a measurement that mutates state rather than reading it purely is a literal observer effect - we have already seen one, where a background thread flipped the model's train/eval flag mid-forward and changed gradients that gate on it; harmonics propagate such a perturbation instead of localizing it. Second, *measurement-induced rounding*: observation forces dtype boundaries a pure forward pass avoids (snapshots cast; the inverse transform is sensitive in its low-order bits), and on the exponential edge of Section 5 rounding can amplify rather than wash out - precisely where the field is most plastic. Third, the run writes records of itself into its own training distribution. At launch the framework renders the model's blueprint to the terminal on a time-ramped curve and stamps the elapsed wall-clock as a rate - a number set by disk caches, scheduler jitter, and thermal state, not by the seed - and rewrites the LICENSE copyright line with the instant of observation expressed as a fractional year (2026.8352091670904681 at this build), which is then committed. Because

the repository is part of the corpus the model trains on, the model ingests the timestamps of its own observations: running the experiment edits the experiment's data. The first two artifacts are testable as a diff between a watched run and an unwatched one from the same seed, against a non-harmonic control that isolates the field's non-locality. The third qualifies what "unwatched" can mean: the two arms can never share execution timing, so the complete control is a deterministic, bit-for-bit replay of the entire input timeline - what speedrunners call a tool-assisted run - and anything short of that bounds the effect rather than isolating it. The strong form is the architecture's own: the remote-expert pool the framework runs is, by construction, a passive observer of the model's batches whose gradients never return, so drawing that single coupling - observer to participant - is a clean two-arm ablation of whether being watched, when watching also feeds back, changes which patterns manifest.

8 Discussion

8.1 Answering the opening question

Do you exist in other people's realities the same way you exist in your own? If the geometry of an attention architecture determines which patterns it constructs, then observers running different geometries build different representations from the same input. You exist in your reality through your geometry, and in another's through theirs. This is not solipsism: convergence to a shared pattern space (Figure 6) says everyone is reaching the same place. It is the *path* that differs - which patterns manifest, in what order, at what cost (Figure 7) - and the path is set by architecture.

The same lens has to account for its harder cases - the readings that are not merely different but look incoherent from outside. A shared document can anchor an elaborate, internally consistent interpretation across a large community and read as noise to everyone else; the declassified Gateway Process and psychotronics assessments in the intelligence archives are a standing example [36, 1], self-evident inside one geometry and opaque from another. The framework neither endorses such a reading nor waves it away - it names it. On the shared field the corpus rhythm is the convergent part, the bias every observer reaches; a construction that does *not* converge to it is *variance* - real, coherent on its own coordinates, and told apart from the signal by its failure to converge rather than by anyone's authority to rule it out. That is the anti-solipsism guardrail above doing its work. We are not forced to accept divergent logics as true; we are forced to accept that they are *constructible* - that a variance-driven geometry can grow arbitrarily elaborate and still be variance. The discipline is to read coherence-to-an-observer as evidence about geometry, never as evidence about the field.

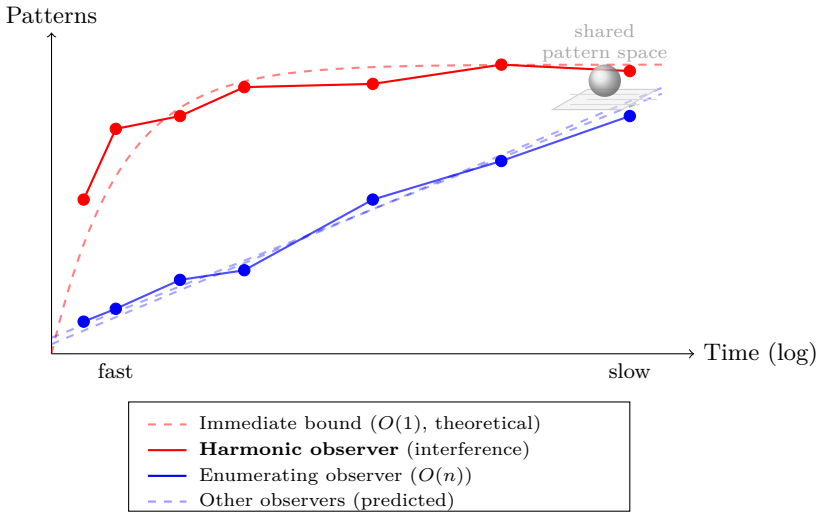


Figure 7: Schematic, not measured. The same convergence as Figure 6, read in the time domain rather than the resource domain. Different observers reach the same shared pattern space (the sphere over the plane, top right) along different temporal complexities: a harmonic observer composes by interference and has near-immediate, $O(1)$-like access (red, tracking the theoretical bound); an enumerating observer accumulates the same patterns one at a time, $O(n)$ (blue), and other architectures (dashed) sit between. All converge asymptotically - the destination is shared, and it is the *path* to it, set by architecture, that differs. This is the intuitive companion to the cost figure, restored from an earlier draft and redrawn as the schematic it always was rather than the measurement it was once dressed as.

8.2 When two observers couple

If a single observer constructs its reality through its geometry, the next question is what happens when two geometries are not merely different but *coupled* - phase-locked to a common rhythm rather than running independently. The harmonic substrate gives a sharp prediction. Two weakly coupled oscillators with nearby natural frequencies lock together, and at the edge of that locking region the system sits near a bifurcation where its response to a small perturbation diverges. A pair tuned to a shared frame is therefore poised to amplify a perturbation of that frame that either alone would miss. Carried to people, this is the everyday report of *feeling watched*: the conjecture is not that there is a new sense, but that coupling is a gain on an old one. The strongest case for a literal channel is adaptive - a prey animal that senses a fixed predatory gaze outlives one that does not, and the payoff is asymmetric enough, a cheap false alarm against a fatal miss, that selection would favor it - but that same asymmetry is also the deflation: error-management logic builds a deliberately biased, hair-trigger detector, fed by a fast subcortical route that tags gaze and looming

motion below awareness, and it predicts both the near-universal *experience* and its poor accuracy, a real and adaptive sense that is heightened ordinary perception rather than a new modality. The folk phenomenon is, on the evidence, best explained away - detection collapses to chance once conventional sensory channels are removed and the looking sequence is properly randomized - but the explaining-away has a regime baked into it that the studies never controlled. They pair strangers, who are uncoupled by construction, and they remove the very channel a coupled pair would amplify. The weak, measurable form follows directly, and it dissociates the two readings: stratify pairs by a measured synchrony index (interpersonal neural and physiological entrainment are now routinely recorded), and cross coupling with sensory isolation. If detection rises with coupling *only* while a conventional channel is open, and vanishes under isolation even for the most synchronized pairs, then the effect is a resonance gain on ordinary cues - the same diverging susceptibility this paper invokes near criticality elsewhere - and nothing exotic is required; if it instead survives isolation, the claim is extraordinary and belongs to a pre-registered replication, not to this paragraph. As with the model's observer effect (Section 7), the quantum reading is metaphor we do not cash, and the engaged-versus-passive distinction is the same one the framework already runs: a watcher coupled into the frame is the human face of drawing the passive observer pool back into the forward.

8.3 A note on reconciliation

The same convergence principle drives the author's companion infrastructure project, Platformer [7], where the explicit design stance is eventual consistency: rather than seeking perfect synchronization, the system runs repeated reconciliation cycles that drift toward a declared desired state and "find balance" through iteration. The shift there from fragmented to unified state is described in the same terms as the shift from sparse to fully-connected networks - more global context yields better optimization. Training is reconciliation of this kind: a model is run against its objective again and again until it settles into an attractor, the way a harmonic field settles onto a corpus rhythm or a codec phase-locks. Platformer carries the dual, too: its disaster recovery is *sequence-modeled*, lifecycle events forming a convergence graph walked *forward* to deploy and *backward* to revert - recovery as traversal of a reversible graph rather than restore from a snapshot. That is the harmonic pass's own shape at another scale: the hidden state wound forward into a geometry and released back to the basis (Section 8.5), development and reversion as two directions on one graph. That the same philosophy organizes both a language model and a fleet of cloud accounts is not a coincidence of the author's taste; it is what convergence under repeated local correction looks like, in whatever substrate.

Carried to its edge, the same thought reaches an intuition we will name but not bank: that a mind may be coupled to the world it runs inside as directly as a model is to its corpus - energetically, not only informationally. The only form of it we defend is the one already staked out under coupling (Section 8.2), a

measurable resonance gain on ordinary channels near criticality; the stronger, non-local reading is left to the experiment that would earn it, not asserted here. The proof, here as everywhere in this framework, would be in the doing.

8.4 A structural remark on temperament

The orthogonal-axes reading (Section 3) has one application worth stating carefully, because the domain invites exactly the misreading the framework is built to refuse. Two dispositions toward a body of accumulated structure - preserve what has been tested against the world, or explore configurations the tested structure has not yet reached - map onto bias and variance directly: the first is high-bias, and its excess is ossification, dogma, and the mistaking of a surviving consensus for the whole truth; the second is high-variance, and its excess is instability, the discarding of tested structure before anything has replaced it. Read as a single dial, this is the oldest political cliche, and it is partisan by construction, because a dial forces a choice of which end is "more." Read as orthogonal coordinates - the paper's actual claim - the dial dissolves: a system, whether a model or a polity, wants low error on both axes at once, and the failure mode is not either disposition but its excess left unopposed - a monoculture on either pole. This is a claim about structure, not about any concrete question, party, or era, and it says nothing about who is right on any of them; it says only that treating preservation and adaptation as one quantity to trade off is the same mistake this paper spends its middle sections arguing against in the model, wherever else the mistake recurs.

8.5 Blind watchmaking, made literal

The title turns Dawkins' figure into an activity, and it should earn the borrowing rather than lean on it. His argument is that apparent design needs no designer: *cumulative* selection - small variation retained and built upon, step after step, with no foresight - reaches structure that single-step chance never could [11]. His *biomorphs* make it concrete - a handful of recursive developmental rules, treated as genes, grow into branching tree-like forms that evolve under selection, the genes never coding the form directly but coding the *development* that grows it. Read the harmonic model in those terms and the metaphor stops being decoration. The frozen basis is the genotype: the inherited, stable rule, cut once by the Weyl seeding and never re-cut. The construction of the field and the contortion of the hidden state across recurrent-depth steps are the development - the rule winding up into a phenotype, geometries stretched somewhere else for a moment, branches growing, roots crawling, strands entangling, before the pass resolves. The vote and the halting convergence are selection; the loop and the repeated passes are what make selection cumulative rather than single-step; and that no one specifies the spectrum the field locks onto, that it emerges and we only read it off afterward (Section 8.6), is what makes the watchmaking blind.

The gerund is the whole of the difference, and it is a correction rather than a

flourish. Dawkins' point is that there *is* no watchmaker, so a title naming the agent quietly reinstates the thing the argument removes; naming the activity does not. It is also the more honest description of what this framework does. We are not presenting a watch and asking to be believed about how it was made. We are building them - many of them, in configurations whose outcomes we do not know in advance, keeping what survives contact with the data and discarding what does not - and the paper is a report from inside that process rather than a verdict delivered after it. Section 8.6 makes the same point about proof; the title makes it about the work.

Set both arguments on the stage, because this is a place they can be told apart. Dawkins' gradualism does not demand a constant rate - only that complex structure accrue through small cumulative steps rather than single saltational leaps; fast bursts are allowed, sudden whole-form jumps are not [11]. Eldredge and Gould's *punctuated equilibrium* pressed the opposite emphasis: that the dominant tempo is long *stasis* interrupted by geologically brief change [14]. Much of that dispute turned on the fossil record itself - are the gaps between forms missing intermediates, as gradualism reads them, or real stasis, as punctuationism reads them? - and it could not be closed, because the record is incomplete by construction: most individuals never fossilize. A forward pass has no such gap. Every computational step can be snapshotted, the developmental record made complete, and the question paleontology could only litigate becomes one a measurement settles.

That is what gives the title a measurement and not just a mood. If the contortion is a *wind-up that releases back to the basis* rather than a drift away from it, then across steps the state should be *stable* - close to the standing harmonic shape - punctuated by *sparse, discrete jumps* in magnitude where a geometry is briefly expressed and then reverts: the punctuated tempo, not the smooth one. The conjecture this paper registers takes that side, and it cuts both ways - a state that drifts continuously and never returns to the basis would refute it, and so would one that changes by a little at every step, which is the gradualist reading made visible. It converges with the always-grokking conjecture, which predicts the same punctuation read as which symbol is active rather than as magnitude, and the measurement is laid out in the project notes. And the deepest part of Dawkins' objection stays in reach: whether a given jump is a true saltation or the visible crest of many small accumulated moves is itself in the snapshot, so the model need not simply side with Gould - it can show which of the two each step was. That is the title made literal: development under blind, cumulative selection, snapped step by step, reverting each time to the inherited rhythm it was grown from.

8.6 The practitioner as proof

That blind, cumulative process is what these systems run. No one specifies the spectrum the harmonic field will lock onto, or the geometry the crystal centers will grow into; they emerge from the process, and we read them off afterward.

Which raises an honest question about what would even count as proof here. A benchmark number proves that one model beat another on one distribution. It does not establish a claim about regimes of representation. The framework's proof, if it has one, is of a different kind: it is the practice itself. *Praxis* is the Greek word for theory enacted - for doing as a form of knowing. The demonstration on offer is not a leaderboard but a method you can run, whose intuitions arrive only by stepping through it yourself. The argument is in the name: the proof is in the doing.

8.7 Implications for machine learning

If bias and variance are separable in a harmonic latent space, then they should be learned as input-dependent functions rather than chosen as a global operating point: the field learns which rhythm matters, the delta learns when and how much to deviate, and no per-experiment hyperparameter mediates the two. If the scaling conjecture holds even partially, then on problems with strong periodic or compositional structure, the right move is to change the basis rather than to add parameters, and the linear and logarithmic regimes should be distinguishable by measuring the slope of coverage against resources. Both are testable in the framework.

8.8 Why architecture is not neutral ground

A standing objection deserves a direct answer, because the framework's whole premise stands or falls on it. If universal approximation [27] guarantees that a sufficiently wide network of *any* family can represent any function we care about, then any Praxis experiment - however intricate - is by hypothesis emulable by some other network, and one is entitled to ask where the value of the architectural complexity could possibly live. The answer is that the objection conflates three distinct equivalences, and universal approximation establishes only the first of them.

Representational equivalence says the function exists in the closure of each family; it is an existence statement at unbounded width. Learnability equivalence would say stochastic gradient descent, from a random initialization, actually *reaches* that function in feasible compute and data. Statistical equivalence would say two families generalize equally at a fixed sample budget. Universal approximation asserts the first and is silent on the other two by construction - it quantifies existentially over parameters, not over what an optimizer finds or what a finite sample supports. The value of an architecture lives entirely in the second and third layers, which is exactly where the existence theorem says nothing.

The foundational mathematics here cuts *for* the framework, not against it. Depth-separation results (Proposition 2) exhibit functions whose representation cost differs exponentially between architectures that are equal as sets of representable functions: the family is the same in the limit, the resource at fixed

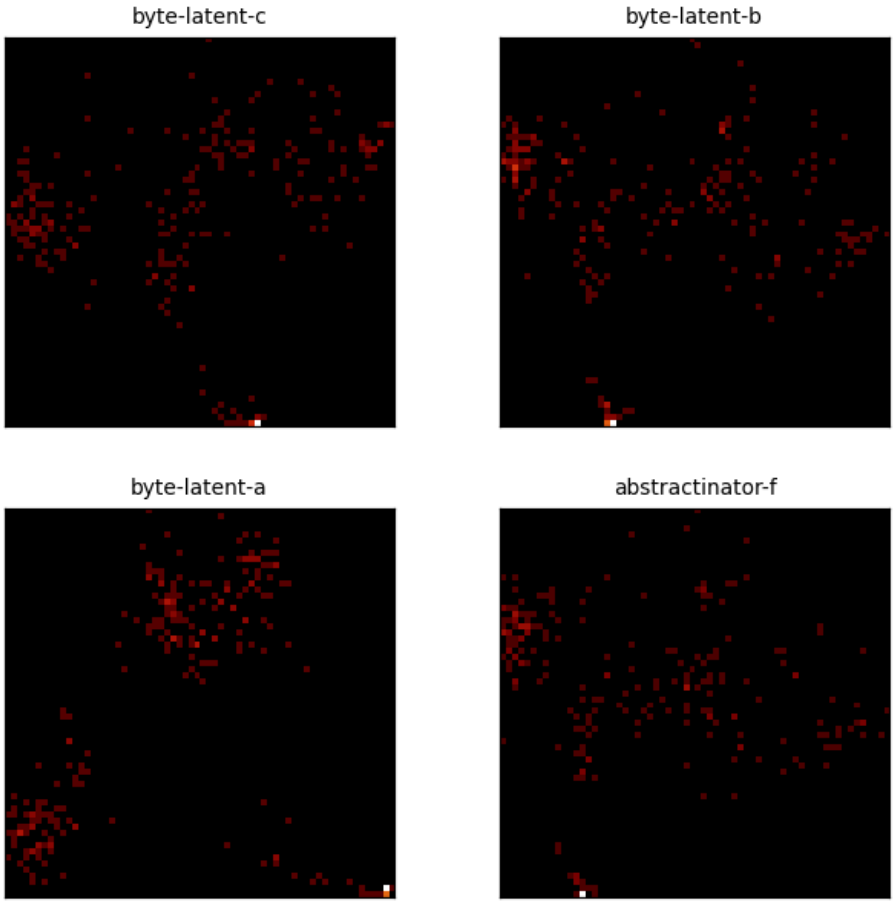


Figure 8: Center PCA density for the 4 most recent crystal-head runs (byte-latent-c, byte-latent-b, byte-latent-a, abstractinator-f). Each panel is the top-2 PCA projection of a head's vocabulary centers, binned to a 64×64 density grid (log color) - the same snapshot the dashboard renders live. Different runs settle into structurally different geometries, not noise.

accuracy is separated by $2^{\Omega(k)}$. "Architecture is just a constant factor" is therefore not a conservative default but a claim already contradicted by theory; the burden of proof runs the other way. The honest residue is that representation cost is not yet learnability, and proving that a given pair of comparable-capacity architectures have *separated error boundaries under training* - the empirical edge of the conjecture in Figure 6 - requires controlled ablations we have not yet run. We state the hypothesis before we have closed it, and mark the seam plainly: the separation of representable cost is established mathematics; the separation of *learned* error across configurations of equal budget is the measurable prediction the framework exists to test, not a result it already owns.

Proposition 2. *Let $\mathcal{F}_A$ denote the set of functions a family of networks with architecture A can approximate to arbitrary accuracy given unbounded width. Universal approximation [27] asserts $\mathcal{F}_A = \mathcal{F}_B$ for any two such families: as* sets of representable functions *they are equal. This equality does not imply equality of* cost *- the width, depth, or sample budget at which a target is reached - and there exist target functions for which the cost gap between two architectures is exponential.*

Proof. The first claim is immediate: universal approximation quantifies existentially over parameters at unbounded width, so it constrains only which functions lie in the closure of each family, not the resources at which they are reached. Equality of sets is compatible with any cost profile whatsoever.

The separation is witnessed, not asserted. Telgarsky [54] exhibits functions on $[0, 1]$ that a network of depth $\Theta(k^2)$ and constant width represents exactly, but that any network of depth $O(k)$ requires $2^{\Omega(k)}$ width to approximate. Eldan and Shamir [13] exhibit a function expressible by a depth-three network of polynomial width that any depth-two network must use exponential width to approximate. In both, $\mathcal{F}_A = \mathcal{F}_B$ holds in the unbounded limit while the resource at fixed accuracy is separated exponentially. Hence representational equivalence is the wrong invariant for comparing architectures of bounded size; the relevant quantity is cost, and cost is architecture- dependent by these results. The further descent - from representation cost to what stochastic gradient descent actually *reaches* from random initialization, and to what *generalizes* at a fixed sample budget - is the inductive-bias question, on which approximation theory is silent by construction; the lottery-ticket phenomenon [16] is one empirical marker that two equal-capacity families do not induce the same reachable solutions. We import these as the foundation the framework's conjecture stands on, not as results re-derived here. □

8.9 Neither halting nor blocking

A recurring objection holds that a neural network cannot be Turing-complete, and the halting problem is offered as the reason. The invocation is a category error. Turing-completeness is the capacity to express any computation, including ones that never halt; deciding in advance whether an arbitrary program halts

is a separate question, and an undecidable one for Turing machines themselves. The theory in fact runs the other way - recurrent networks with unbounded precision and unbounded time are already Turing-complete [46]. What bounds a real model is not undecidability but finite precision and a finite number of steps, exactly the resources a fixed-depth forward pass exhausts and exactly the ones recurrence and delay restore.

Praxis leans into that. With the swarm online, computation runs in two directions at once. Forward, the recurrent loop propels generation and is never required to emit a stop: the halting gate is a convergence check, not an exit, so generation is a continuous autonomous stream rather than a program that terminates. Sideways, a pool of remote experts votes - stochastically sampled, mixed by an expert vote in the style of CALM's patch vote [44] - and trains without blocking, so a slow peer never stalls the others; its update simply lags and lands on a later pass. Computation is lagged as much as propelled, and the model commits to neither a halt nor a barrier. It approximates both by doing neither.

This relocates the question. It is no longer whether the network halts but what the pool expresses, over how much time, at what precision - the resources, not the stop bit. The butterfly is the right image: one expert, then two, then four, each a wing whose small vote the mixture amplifies or cancels, the whole field sensitive to where the late votes fall. The interior turns of the recurrent loop (Section 5) already carry the decision; the last is only the budget running out. Power here is bought with peers and patience, not with a proof that the machine knows when to stop.

The same shape is built into the interface, where it can be used rather than only argued. A network is Turing-complete only with constraints applied - finite precision, a finite step budget, and some structure on what it may do - and Praxis makes one such constraint visible as the Gymnasium, a read-eval-print loop turned from a sequence into a choice. A classical REPL runs its four stages in series - read, then evaluate, then print, then loop - each blocking on the one before. Praxis treats them not as stages but as actions and activates one at a time: *Read* the knowledge base, *Evaluate* a conversation, *Print* a question of its own, with the recurrent loop the implicit fourth that carries the stream onward - an XOR over the handful where the conventional loop is their dense, ordered product. Sparse, mutually exclusive action selection is the halting-and-blocking argument in miniature: a loop that neither marches a fixed sequence to a terminating print nor blocks at each stage, but picks a single move and continues. Read and Evaluate are live; Print is the planned next button on the row, and the constraint is the point - the machine is general because of what it is forbidden to do all at once, not in spite of it.

A living document can also look at itself. Praxis is built in the open, commit by commit, and that history is just another sequence - a stream of byte-level diffs over time. Figure 9 turns the framework's own recency kernel on its repository: per-subsystem churn across the project's history, with the distant past faded

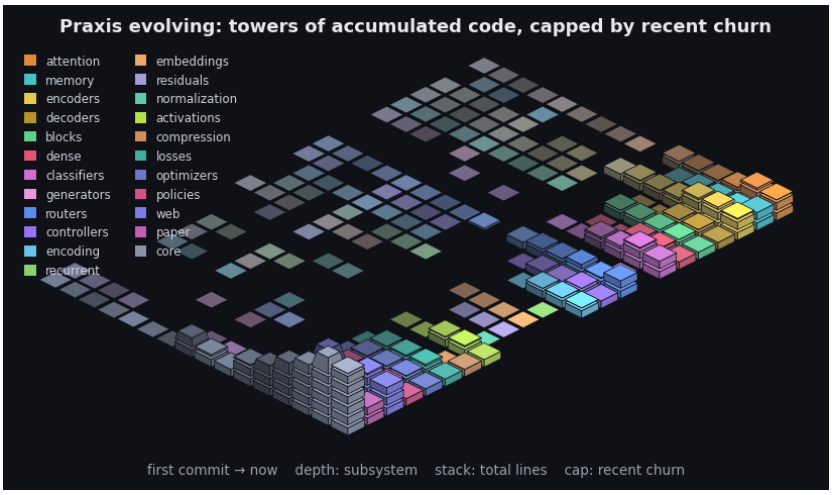


Figure 9: Praxis's git history as an isometric terrain over the last 2702 commits. Each cell is one subsystem in one time window: a stack of base blocks for its accumulated size (total lines, recency-weighted) topped by a vivid cap for that window's line churn - the standing codebase carrying the recent activity. Both are weighted by recency (the kernel the model applies to a sequence, turned on the repository): recent windows stack tall with bright caps while history thins toward the always-present prior - the low valley we roll into. Color fades from each subsystem's hue toward a neutral prior into the past, banded in phased strata over a non-linear timescale. It is the bias/variance geometry read along the frequency axis (see the project notes): loud recent corrections over the quiet, ever-present base. By recency-weighted churn the current center of gravity is core, web, encoders. The framework charts the hand that builds it.

in proportion to its distance from the present, the way a decay bias weights earlier positions in a sequence. It is not an argument, only a mirror - the same machinery that reads a corpus, reading the record of its own construction. The reading that any finite byte sequence is a position to be classified the same way, the repository included, is left as a conjecture to test rather than a claim to lean on (see the project notes).

8.10 Standing conjectures

We gather the paper's falsifiable conjectures in one place; each is discussed where it arises. They are bets about reachability and cost, not theorems, and each names what would refute it.

- **Linear consensus, logarithmic interference.** Learning a corpus as a consensus mean costs resources linearly in the patterns expressed, while composing it by harmonic interference costs only logarithmically, so the coverage-versus-resources slope a basis buys is measurable and basis-dependent.

The representational half is no longer conjectural - the amplitude-to-field map is an isometry, so a K-dimensional resolution grid carries to exponentially many separated fields - which leaves two things to measure: whether gradient descent reaches those configurations, and whether the spectrum stays spread enough to hold them, since capacity saturates in K under a fixed envelope with bounded amplitudes. [28]

- **Bias and variance as orthogonal axes.** Bias and variance - the projections of the harmonic latent onto its frozen, time-invariant modes and its time-varying modes - occupy orthogonal subspaces whose loss cross-curvature vanishes, so under training both fall together toward the origin rather than trading off along a frontier; the claim is falsified the moment the measured (bias, variance) trajectory bends into a Pareto frontier, or the cross-curvature between the two mode bands stays bounded away from zero. Orthogonality alone is not sufficient: by the interference-capacity proposition the variance axis carries configurations only while the input-conditional spectrum stays spread, so a delta whose amplitudes concentrate alongside the static baseline - or decay with the $1/f^{\alpha}$ envelope instead of resisting it - leaves the axes orthogonal and the second one empty, which refutes the useful form of the claim.
- **Always grokking.** Because the grokked solution is itself a harmonic representation, a model with frozen phases begins post-grok and never undergoes the delayed memorize-then-generalize transition over training; the transition instead recurs on the input axis - the symbol set is fixed, but the belief state is perpetually re-fitting, hopping between those fixed symbol clusters across compute steps - and the claim is falsified the moment that per-input path is measured to be a smooth drift rather than discrete cluster-to-cluster hops, or the symbol set is found to keep growing rather than being recombined. [39, 43]
- **Development reverts to the basis.** If the contortion of the hidden state across recurrent-depth steps is a wind-up that releases back to the frozen harmonic basis - development expressed and then reverted, in the manner of cumulative selection - then the per-step state trajectory should be stable, close to the standing harmonic shape, punctuated by sparse discrete jumps in magnitude rather than a continuous drift, and the claim is falsified the moment that trajectory is measured to drift smoothly and monotonically away from the basis without returning, or to change at every step rather than in rare punctuated events. [11, 14]
- **Information density at the rim.** Information density - the share of the representation's distinguishing content carried at each position - is uniform by assumption in the consensus regime but pushed to the window's extremes by a harmonic latent, a slow hum anchored at the head and an input-conditional chirp at the tip, so hidden-state deviation must grow with recurrent depth under a monotone positional gradient - early positions settling toward a fixed point while the tip stays live - wherever a re-

currence is harmonized; the claim is falsified the moment the per-position, per-step deviation profile - read in norm and in symbol occupancy alike, since a silent bit flip can change the geometry without moving the norm - is measured flat in position, or fails to steepen across depth. [61, 50]

- **Reachable, not merely representable.** Across architectures of comparable parameter count and compute there exist task and data regimes where generalization error is separated, because the prior sets which solutions stochastic gradient descent can reach, not only which functions exist in the closure. [54, 16]
- **Lottery engineering.** A geometric prior makes good sparse subnetworks reachable by stochastic gradient descent directly rather than only by train-then-prune, and a rotating population of such subnetworks (mixture-of-widths) approaches full-width performance at lower average active width. [16]

8.11 Limitations

We are deliberate about what is and is not established. The bias-variance decoupling is grounded in a running implementation and its diagnostics (concentration, smoothness, delta norm, the spectrum heatmap), but "orthogonal axes" is an interpretation of those diagnostics, not a theorem. The scaling conjecture of Figure 6 is a conjecture: the figure draws an asserted relationship, not measured curves, and the honest next step is to measure the slope on tasks of varying combinatorial structure and report where harmonic representations fail to buy it. The observer-relative view of tractability (Section 5) is a philosophical position, and only its weak form - the scaling slope a given basis buys on a given problem family - is measurable; we make no claim about the formal worst-case P versus NP question, which abstracts the observer away by design. The connection to biological attention remains speculative, and the prismatic asymmetry needs to be shown in silico now that the hand-measured figure is gone. The observer effect of Section 7 is, in its quantum form, an analogy we do not cash; only its weak form is a claim - that impure measurement (state-mutating reads, measurement-induced rounding on the exponential edge, self-ingested execution timestamps) perturbs a non-local harmonic field measurably more than an independent-feature one, and that coupling the framework's passive observer pool back into the forward is a two-arm ablation of being-watched. The temporal artifact in particular is a claim about data provenance, not physics: launch rates and license stamps enter the model as tokens, so "modeling its own time" means modeling those tokens, and we make no claim that the field encodes execution timing beyond what is literally written into its corpus. Those are diffs between a watched and an unwatched run, not statements about physics. The coupled-observer extension (Section 8.2) - that two phase-locked observers amplify perturbations of the frame they share, of which *feeling watched* is the folk instance - is conjecture in the same spirit: its only measurable form is the synchrony-stratified design that separates a resonance gain on ordinary sensory cues from a non-sensory channel, and we side explicitly with the former, against

the latter, pending that experiment. None of these are reasons not to state the ideas; they are the boundary between what we have built and what we are conjecturing, drawn so the reader can attack the right target.

8.12 A direction: the moving point

One reading the geometry keeps inviting is worth stating as we close, marked as a direction rather than a result. The case for the third state was that two points fix only a line and three fix a plane (Lemma 1), and that a phase can turn only in a plane - so the least representation that can hold an oriented shape is two-dimensional, not one. But that plane is only where a phase *sits*. A phase also moves: it carries a velocity and a direction, and a position taken together with its motion is a phase space, twice the dimension of the configuration it lives on. The line becomes a plane, and the plane the harmonic field was built to turn in becomes the four-dimensional space in which that turning is itself a coordinate. The representation we have described is therefore never a point; it is a moving point, and the climb - from a position, to a line, to the plane where phase lives, to the phase space where phase moves - is the same ascent at every scale. We make no claim that the network parameterizes that space explicitly. We note only that the geometry, followed honestly, does not stop at the plane: velocity and direction, not position alone, are what a harmonic representation is finally about.

Lemma 1. *Two distinct points determine a line; three non-collinear points determine a unique plane. A phase - a point on the circle S^1 - is realizable only in dimension at least two, so three points are the minimal affine data that orient the plane a harmonic component turns in.*

Proof. Through p_0, p_1, p_2 the affine span $\{\, p_0 + s(p_1 - p_0) + t(p_2 - p_0) \,\}$ is a plane exactly when $p_1 - p_0$ and $p_2 - p_0$ are linearly independent - that is, when the three points are non-collinear; two points give a single direction and hence only a line. Since S^1 embeds in $\mathbb{R}^2$ but in no line, a continuous phase requires that plane, and three points are the fewest that fix it.

9 Conclusion

We began with scale and ended with geometry. The patterns a model can learn are set by the shape of the space its weights may occupy, and a harmonic latent space reshapes the oldest tradeoff in the field: bias becomes a static, population-average spectrum, variance becomes an input-conditional modulation, and the paper's structural claim is that they live on orthogonal axes and can therefore be lowered together. From that decoupling follows a conjecture stated plainly - that learning the mean costs linearly while composing by interference costs logarithmically - and a framework built to test it rather than to argue it. And from that, in turn, follows the harder claim: that tractability is not a fixed property of a problem but a relation between a problem and the observer solving it. The

same instance is exponential to one architecture and nearly free to another, so P and NP are better read as a spectrum that the choice of representation indexes than as a wall every solver meets in the same place. The speculative figure is gone; the conjectures that replace it are falsifiable. What remains is praxis: the answer manifests through direct experimentation, and what you choose to attend to determines what we find next.

A Current Experiment

This appendix is generated. The figure is built from training logs by the framework's own paper build; the paper itself is a template that build populates. The chart leads with the most recently launched experiment that has logged validation data, and fixes the y-metric to that run's family - here Bits per Byte, as of the last export (2026-11-01). Only experiments that report the *same* metric are drawn alongside it; the rest are listed as omitted below. This is the smallest living slice of the document: one metric, the current experiment against its compatible predecessors.

Note:

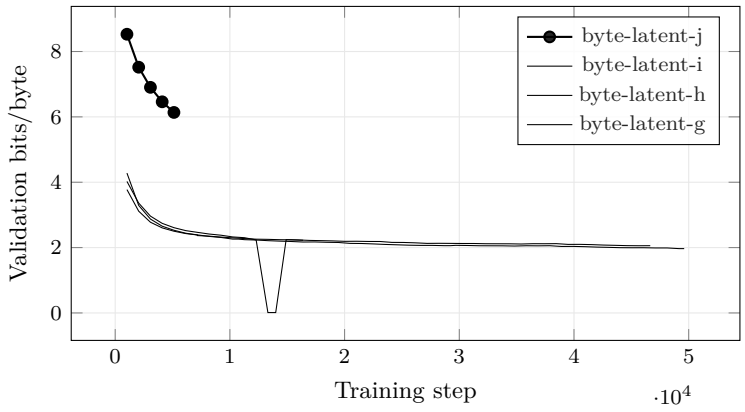


Figure 10: Bits per Byte across the 4 most recent experiments with comparable validation curves; the current experiment (byte-latent-j) is drawn in bold. Bold marks *which run this build is reporting on*, not which run scores best. The curves end wherever each run stopped, so they are at different points along their own training, and their endpoints are not a ranking. Generated from run logs - metrics that are not comparable across families (token loss vs. byte bits-per-byte vs. codec/CALM BrierLM) are excluded by construction, so the curves on this axis mean the same thing.

The current experiment, byte-latent-j (`9f8a6751d`), reached Bits per Byte 6.136 by step 5121 - the value at the point this build was taken, not a converged result, and not a claim against the other curves, which stopped at step counts of their own. The structural figures elsewhere in this paper are read from this same

run, so they are likewise a snapshot of a run in progress rather than of a settled model; that is what makes them a live reading and also what limits what they can be asked to prove. Recent experiments whose families report a different validation metric are omitted rather than plotted on an incomparable axis: .

Dataset	Type	byte-latent-j	byte-latent-i	byte-latent-h	byte-latent-g
praxis	directory	1	1	1	1
synthetic-tool-calling	synthetic	1	1	1	1
fineweb-edu-350bt	huggingface	1	1	1	1
fineweb	huggingface	1	1	1	1
finepdfs	huggingface	1	1	1	1
persona-chat	huggingface	1	1	1	1
wildchat	huggingface	1	1	1	1
natural-instructions	huggingface	1	1	1	1
cosmopedia-v2	huggingface	1	1	1	1
smoltalk	huggingface	1	1	1	1
git-history	git_history	1	1	1	1
platformer	directory	1	1	1	1
rift	directory	1	1	1	1
vtx	directory	1	1	1	1
src.eco	directory	1	1	1	1
eliza	directory	1	1	1	1
fold	directory	1	1	1	1
ode	directory	1	1	1	1
dimanet	directory	1	1	1	1
fvn	directory	1	1	1	1
ext	directory	1	1	1	1
ink.university	directory	1	1	1	1
l.ink	directory	1	1	1	1
reaper	directory	1	1	1	1
cv	directory	1	1	1	1

Table 1: Training-data mixtures behind the 4 most recent experiments, resolved from each run's recorded configuration: named collections expand to their member datasets, and every explicitly configured data directory is its own sampler. Cells are the configured per-dataset sampling weights (pre-normalization), not token counts; – marks a dataset absent from that run's mixture.

References

[1] The psychotronic universe. https://www.cia.gov/readingroom/docs/CIA-RDP96-00792R000400330013-4.pdf, n.d. Declassified, CIA Reading Room record CIA-RDP96-00792R000400330013-4.

[2] Anonymous. Is random attention sufficient for sequence modeling? disentangling trainable components in the transformer. arXiv:2506.01115, 2026.

[3] David D. Baek, Ziming Liu, Riya Tyagi, and Max Tegmark. Harmonic loss trains interpretable ai models. *arXiv preprint arXiv:2502.01628*, 2025.

[4] Jonathan Basile. Library of Babel. https://libraryofbabel.info/, 2015. Algorithmic realization of Borges' Library of Babel: each page returned by a reversible address-to-content function rather than stored.

[5] Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. *arXiv preprint arXiv:2501.00663*, 2024.

[6] Marc G. Bellemare, Will Dabney, and Rémi Munos. A distributional perspective on reinforcement learning. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *PMLR*, pages 449–458, 2017.

[7] Ryan J. Brooks. Platformer: Autonomous systematic mechanical reconciliation. https://github.com/0-5788719150923125/platformer, 2025. Companion project to Praxis; infrastructure-as-code framework with eventual-consistency reconciliation.

[8] Ryan J. Brooks. Praxis: A framework for architectural diversity in attention mechanisms. https://github.com/0-5788719150923125/praxis, 2025. Accessed: 2025-10-11.

[9] Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and brain sciences*, 36(3):181–204, 2013.

[10] Stephen A Cook. The complexity of theorem-proving procedures. In *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, pages 151–158, 1971.

[11] Richard Dawkins. *The Blind Watchmaker*. W. W. Norton & Company, 1986.

[12] Aaron Defazio, Xingyu Alice Yang, Harsh Mehta, Konstantin Mishchenko, Ahmed Khaled, and Ashok Cutkosky. The road less scheduled. *arXiv preprint arXiv:2405.15682*, 2024.

[13] Ronen Eldan and Ohad Shamir. The power of depth for feedforward neural networks. In *Conference on Learning Theory (COLT)*, pages 907–940, 2016.

[14] Niles Eldredge and Stephen Jay Gould. Punctuated equilibria: an alternative to phyletic gradualism. In Thomas J. M. Schopf, editor, *Models in Paleobiology*, pages 82–115. Freeman, Cooper, 1972.

[15] Jakob Foerster. Nonlinear computation in deep linear networks. OpenAI blog, September 2017. https://openai.com/index/nonlinear-computation-in-deep-linear-networks/.

[16] Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. *International Conference on Learning Representations (ICLR)*, 2019.

[17] Josh Freckleton. neurallambda: Fully differentiable lambda calculus. https://github.com/neurallambda/neurallambda, 2024.

[18] Karl Friston. The free-energy principle: a unified brain theory? *Nature reviews neuroscience*, 11(2):127–138, 2010.

[19] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. *arXiv preprint arXiv:2502.05171*, 2025.

[20] Joe Germany, Elie Abdo, and Joseph Bakarji. Eml trees are universal approximators. *arXiv preprint arXiv:2606.23179*, 2026.

[21] Fabian Gloeckle, Badr Youbi Idrissi, Baptiste Rozière, David Lopez-Paz, and Gabriel Synnaeve. Better & faster large language models via multi-token prediction. *arXiv preprint arXiv:2404.19737*, 2024.

[22] Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. *SIAM Journal on Computing*, 18(1):186–208, 1989.

[23] Aidan N Gomez, Mengye Ren, Raquel Urtasun, and Roger B Grosse. The reversible residual network: Backpropagation without storing activations. In *Advances in neural information processing systems*, volume 30, 2017.

[24] Alex Graves, Greg Wayne, and Ivo Danihelka. Neural turing machines. *arXiv preprint arXiv:1410.5401*, 2014.

[25] Werner Heisenberg. Über den anschaulichen inhalt der quantentheoretischen kinematik und mechanik. *Zeitschrift für Physik*, 43(3-4):172–198, 1927.

[26] Geoffrey Hinton. The forward-forward algorithm: Some preliminary investigations. *arXiv preprint arXiv:2212.13345*, 2022.

[27] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.

[28] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[29] Sabine Kastner and Leslie G Ungerleider. Mechanisms of visual attention in the human cortex. *Annual review of neuroscience*, 23(1):315–341, 2000.

[30] Seyed Mehran Kazemi, Rishab Goel, Sepehr Eghbali, Janahan Ramanan, Jaspreet Sahota, Sanjay Thakur, Stella Wu, Cathal Smyth, Pascal Poupart, and Marcus Brubaker. Time2vec: Learning a vector representation of time. *arXiv preprint arXiv:1907.05321*, 2019.

[31] Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.

[32] James Lee-Thorp, Joshua Ainslie, Ilya Eckstein, and Santiago Ontanon. FNet: Mixing tokens with fourier transforms. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics*, 2022. arXiv:2105.03824.

[33] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, 2023.

[34] Yihao Liu et al. Mono-forward: Backpropagation-free algorithm for efficient neural network training harnessing local errors. *arXiv preprint arXiv:2501.09238*, 2025.

[35] Xuezhe Ma, Chunting Zhou, Xiang Kong, Junxian He, Liangke Gui, Graham Neubig, Jonathan May, and Luke Zettlemoyer. Mega: Moving average equipped gated attention. In *International Conference on Learning Representations*, 2023. arXiv:2209.10655.

[36] Wayne M. McDonnell. Analysis and assessment of the Gateway process. https://www.cia.gov/readingroom/docs/CIA-RDP96-00788R001700210016-5.pdf, 1983. U.S. Army Intelligence and Security Command (INSCOM); declassified, CIA Reading Room record CIA-RDP96-00788R001700210016-5.

[37] Igor Mezić. Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41(1–3):309–325, 2005.

[38] Evan Miller. Attention is off by one. https://www.evanmiller.org/attention-is-off-by-one.html, 2023.

[39] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.

[40] OilProducts. Abstractinator. https://github.com/OilProducts/abstractinator, 2025. Byte-latent encoder with a multi-stage residual vector-quantization bottleneck.

[41] Artidoro Pagnoni et al. Byte latent transformer: Patches scale better than tokens. *arXiv preprint arXiv:2412.09871*, 2024.

[42] Raphael Pisoni. HALO. https://github.com/4rtemi5/halo, 2026. Reference implementation. HALO = Hyperspherical Alignment & Latent Optimization, per the companion post "Soap Bubbles and Attention Sinks: The Theory and History of the HALO-Loss", https://pisoni.ai/posts/halo/.

[43] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. In *ICLR MATH-AI Workshop*, 2022.

[44] Chenze Shao et al. Continuous autoregressive language models. *arXiv preprint arXiv:2510.27688*, 2025.

[45] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

[46] Hava T. Siegelmann and Eduardo D. Sontag. On the computational power of neural nets. *Journal of Computer and System Sciences*, 50(1):132–150, 1995.

[47] Ilya M. Sobol'. Sensitivity estimates for nonlinear mathematical models. *Mathematical Modelling and Computational Experiments*, 1(4):407–414, 1993.

[48] Kenneth O. Stanley and Risto Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2):99–127, 2002.

[49] Yixuan Su, Tian Lan, Yan Wang, Dani Yogatama, Lingpeng Kong, and Nigel Collier. A contrastive framework for neural text generation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[50] Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive activations in large language models. *arXiv preprint arXiv:2402.17762*, 2024.

[51] Floris Takens. Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick 1980*, pages 366–381. Springer, 1981.

[52] Matthew Tancik, Pratul Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[53] Yi Tay, Dara Bahri, Donald Metzler, Da-Cheng Juan, Zhe Zhao, and Che Zheng. Synthesizer: Rethinking self-attention for transformer models. In *Proceedings of the 38th International Conference on Machine Learning*, 2021. arXiv:2005.00743.

[54] Matus Telgarsky. Benefits of depth in neural networks. In *Conference on Learning Theory (COLT)*, pages 1517–1539, 2016.

[55] Jayden Teoh, Manan Tomar, Kwangjun Ahn, Edward S. Hu, Tim Pearce, Pratyusha Sharma, Akshay Krishnamurthy, Riashat Islam, Alex Lamb, and John Langford. Next-latent prediction transformers learn compact world models, 2025.

[56] Ilya Tolstikhin, Neil Houlsby, Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Thomas Unterthiner, Jessica Yung, Andreas Steiner, Daniel Keysers, Jakob Uszkoreit, Mario Lucic, and Alexey Dosovitskiy. MLP-Mixer: An all-MLP architecture for vision. In *Advances in Neural Information Processing Systems*, 2021. arXiv:2105.01601.

[57] Stefan Treue and John HR Maunsell. Attentional modulation of visual motion processing in cortical areas mt and mst. *Nature*, 382(6591):539–541, 1996.

[58] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, volume 30, 2017.

[59] Guilherme Vieira Neto and Marcos Eduardo Valle. Ghost features and spooky transfer learning for hypercomplex-valued neural networks. *arXiv preprint arXiv:2608.07735*, 2026.

[60] Hermann Weyl. Über die gleichverteilung von zahlen mod. eins. *Mathematische Annalen*, 77(3):313–352, 1916.

[61] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations*, 2024.

[62] Aston Zhang, Yi Tay, Shuai Zhang, Alvin Chan, Anh Tuan Luu, Siu Cheung Hui, and Jie Fu. Beyond fully-connected layers with quaternions: Parameterization of hypercomplex multiplications with $1/n$ parameters. In *International Conference on Learning Representations*, 2021.